Course details

Code: BMETE11AP65
Lecturer: András Halbritter
Mid-term tests: -
Grading: oral exam

<20 minutes/ person

First 10 minutes: discussion of the randomly drawn topic (see exam topics on next slides). The focus is on the understanding of the concepts.

If you were absent on <= 5 lectures: you can use the slides or anything you wish both for the preparation and for this part of the exam, and you can also request an online exam, if you wish. In case of more absences, you cannot use the slides, and you have to make the exam in person.

Second 10 minutes: flash questions on any topic. You cannot use anything, but only the most relevant message of each topic is asked.

Grades:
- 5: High level of understanding for the whole material
- 4: Good level of understanding for the whole material
- 3: Significant understanding with significant shortages
- 2: Weak, but still somewhat valuable knowledge

The language of the oral exam is English, but the English knowledge is not an aspect of grading.

Webpage: course webpage in the course team

(Loosely related) literature: -*Keithley Low Level Measurements handbook*
-Varga Dezső és Bagoly Zsolt (ELTE TTK): *Elektronika és méréstechnika*
-James A. Blackburn: *Modern Instrumentation for Scientists and Engineers*
-Fredric J. Harris: *On the use of Windows for Harmonic Analysis with the Discrete Fourier Transform* Proceedings of the IEEE 66 p51–83 (1978)
-Heinzel, G.; Rüdiger, A.; Schilling, R. (2002). *Spectrum and spectral density estimation by the Discrete Fourier transform (DFT), including a comprehensive list of window functions and some new flat-top windows*

1

Exam topics #1-10 (see next slide for #11-20)

1. A motivating experiment for the subject of measurement techniques: demonstrating the zero resistance of a superconductor. Tricks and difficulties in resistance measurement: cable and contact resistance, four-probe measurement method; digital error; nonlinear temperature sensor characteristics; hysteretic cooling and warming. More accurate experiment by an alternative method: persistent current measurement with a magnetic field sensor. The role of bridge configuration in magnetic field measurement.
2. PID regulation. The role of parameters P, I and D, the time constant of the control, problems in case of incorrect setting. Applications: temperature control, atomic distance control (scanning tunnelling microscope, STM), Michelson interferometer.
3. Ideal voltage and current sources, ideal voltage and current meters. Simple voltage and current generator circuits. Operational amplifiers: comparator, non-inverting and inverting voltage amplifier, current amplifier, logarithmic current amplifier and charge amplifier circuits.
4. D/A converter with summing amplifier circuit. Flash and successive approximation A/D converters. Data acquisition card specifications and input settings. Concepts of resolution and accuracy. Common mode rejection.
5. Basic functions of analog and digital oscilloscopes: input coupling, triggering, averaging and enhanced resolution, ultra-fast sampling methods. Measurements with oscilloscope. Synchronization, function generator burst mode. Concept of complex impedance. RC time constant problems due to stray capacitances in high frequency measurements.
6. Wave propagation in coaxial cables, telegrapher's equations. The concept of wave impedance. Reflections at the cable end, the role of impedance matching. Measurement of signal propagation speed by standing wave and time of flight methods. Reflection and transmission on in-line resistors.
7. Suppression of external interference: electrostatic coupling, shielding and grounding; inductive coupling, twisted pairs; low-frequency disturbances caused by high-frequency interference rectification; thermoelectric voltages, offset compensation; concept of guarding.
8. Fourier series, Fourier transform. Fourier integral over a finite time interval. Discrete Fourier Transform (DFT), consequences of the Nyquist-Shannon sampling theorem.
9. Concept and properties of window functions. Spectral leakage, frequency resolution, amplitude accuracy, and the role of normalization. Properties of rectangular, Hanning and flattop windows. Why can we suppress spectral leakage by a window decaying at the sides of the measurement interval?
10. Concept of fast Fourier transform, FFT spectrum analyzers. Further methods of spectrum analysis: heterodyne techniques, hybrid and swept tuning spectrum analyzers.

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Exam topics #11-20

11. The principle of the lock in amplifier. Phase locked loop, PLL synchronization to external reference signal. Measuring a small signal in a noisy environment using a modulation technique. Noise filtering considerations – why should we use a lock-in? Higher harmonic measurements with a lock-in.
12. Experimental definition of noise. The spectral density of noise, the Fourier transform of the current-current correlation function and the Fourier transform of the current fluctuations, and the relationship between these quantities. Spectral density calculation based on DFT.
13. Thermal noise calculation, relation between conductance and noise through the correlation time. Minimum input noise of a current amplifier. Cross correlation measurement technique.
14. Shot noise of independently emitted electrons, electron charge measurement. Equivalent noise bandwidth. Resistance fluctuations, 1/f type noise. Aliasing in noise measurement, antialiasing filter.
15. International System of Units (SI). Old and new definitions of basic units, reasons for the introduction of the new SI system. Historical evolution of time and distance measurement.
16. Atomic clock as a time standard. The quantum metrology triangle: converting voltage to time through the Josephson effect, current to voltage through the quantized Hall effect and current to time by electron pumps. Mass measurement: Watt balance and Avogadro project.
17. Magnetic field sensors: inductive sensors; magnetoresistive sensors (anisotropic magnetoresistance, giant magnetoresistance and spin valve sensors, tunneling magnetoresistance), Hall probes, SQUIDs.
18. Distance and position sensors: inductive sensors, capacitive sensors, touch screen with pressure sensitive pen, laser rangefinders and ultrasonic distance sensors, LIDAR system.
19. Temperature sensors, primary and secondary thermometers. Thermocouples, resistance thermometers, thermistors. Light and electromagnetic radiation sensors: photodiodes, CCD sensors, CMOS active pixel sensors, bolometers. Accelerometers: MEMS accelerometers and gyroscopes, piezoelectric accelerometers.
20. Measurement of radiation and particles. Interactions of ionizing radiation and matter, neutral particle detection. Gas-filled ionization detectors: ion chambers, proportional counters and Geiger-Müller tubes. Scintillation detectors with photoelectron multipliers. Applications. Fundamentals of nuclear electronics, integral and pulse mode detection, energy resolved spectra.

Introduction – Measurement Techniques

What are measurement techniques? Well, it means different things in every profession.

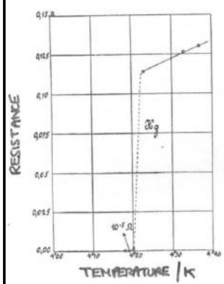
From a physicist's point of view: we usually want to measure a **physical quantity**. Often, a **sensor** with a special physical principle converts that physical quantity into an electronic signal. The accuracy and success of the measurement, however, depends largely on how the sensor signal is processed, what **measurement circuit**, what **measurement setup**, what measurement trick is used. Finally, data are acquired with a precision **measuring instrument**, and various **data evaluation** tricks and data science methods can also be important for processing these data. A measurement technique course may cover almost any of these aspects.

Physical quantity | **Sensor** | **Measurement setup, measuring circuit** | **Measuring instrument** | **Data processing**

Topics covered:
-Measuring superconductivity: an experiment illustrating measurement difficulties and tricks. How could we refine the measurement? The concept of feedback-controlled temperature-regulation: the PID control.
-Modern multimeters, functions generators, DAQ cards, oscilloscopes: the concept of A/D and D/A converters, typical device specifications and tricky measurement modes. Related basics of amplifier circuits.
-Suppression of external disturbance signals: elimination of capacitive and inductive pick-ups by shielding and twisted cables, offset-compensation of thermoelectric voltages, guarding.
-High-frequency measurements: RC-time-constant issues, wave propagation in transmission lines.
-Spectrum analysis, the effect of finite measurement time and sampling rate, window functions, Fast Fourier Transformation (FFT), various types of spectrumanalyzers.
-Phase sensitive detection: lock-in amplifiers and phase-locked loops.
-Natural noise sources in high precision measurements, concept of noise analysis.
-Metrology, the SI system and measurement standards.
-Sensorics: measuring displacement, temperature, light, magnetic field, acceleration, etc.
-Introduction to nuclear measurement techniques

Measuring the zero resistance of a high-temperature superconductor sample – superconductivity in a nutshell

Discovery (1911):
The resistance of mercury drops to zero (unmeasurably small) below T=4.15 K



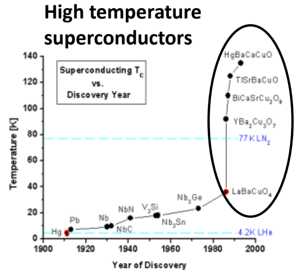
Heike Kamerlingh Onnes

Nobel prize: 1913

Microscopic theory (BCS theory, 1956)

Nobel prize: 1972
Schrieffer Bardeen Cooper

High temperature superconductors



K. Alex Müller and Georg Bednorz

Nobel prize: 1987

Further Nobel prizes:

Landau 1962

Abrikosov and Ginzburg 2003

Josephson and Giaever 1973

Measuring the zero resistance of a high-temperature superconductor sample – resistance measurement

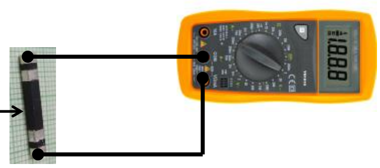
The **resistance** of a superconductor **should drop to zero below a** certain **critical temperature**.
First, we should look up the **sample specifications**: what is the material, what are the dimension, what is the critical temperature, and the maximum allowed current.

CSB-2/3/20 SUPERCONDUCTING BAR
Width: 3 mm
Thickness: 2 mm
Net weight: 0.001 kg
Material: Bi-2223
Length: 20 mm
Max. current: 60mA
Critical temperature: ~108K

How can we measure the resistance of the superconducting rod? Let's try with a digital multimeter!

Physical quantity to be measured: Resistance

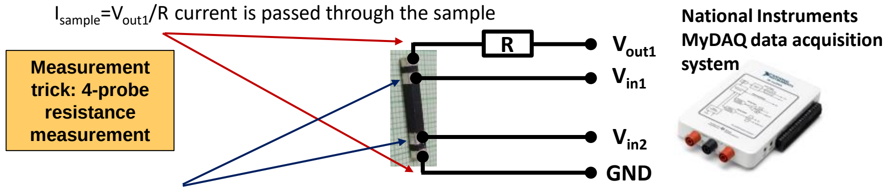
superconducting sample



Do we trust the results? No, we are measuring the resistance of the cables and the contacts in addition to the resistance of the superconducting rod.

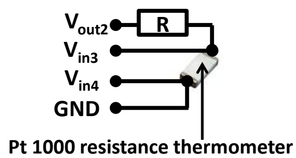
Measuring the zero resistance of a high-temperature superconductor sample – resistance measurement

How can we refine the resistance measurement?



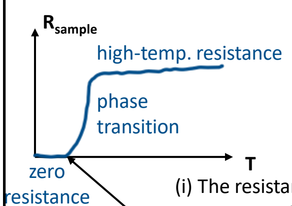
Meanwhile the $V_{sample} = V_{in1}-V_{in2}$ voltage drop is measured on two different contacts from the current injection contacts. The voltage measurement is performed by a voltmeter of almost infinite internal resistance, i.e. practically no current flows in these voltage measurements lines, and no voltage drops on these cables and contacts. Therefore, V_{sample} only includes the true voltage drop on the superconducting rod, and $R_{sample}=V_{sample}/I_{sample}$ is the true sample resistance.

On the other side of the sample holder the temperature is measured by a Pt 1000 resistance thermometer in a “quasi four probe” fashion, meaning that there are only two contacts, but four wires, i.e. the resistances of the voltage measurement wires is cancelled.

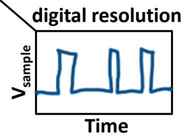


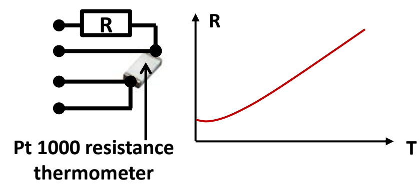
Measuring the zero resistance of a high-temperature superconductor sample – resistance measurement

Let’s perform the measurement. We do see the superconducting transition, but what kind of difficulties do we face?



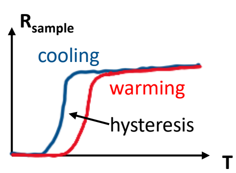
(i) The resistance resolution of the measurement is restricted by the **digital resolution** of the card





(ii) Due to the thermometer’s **nonlinear resistance vs. temperature characteristics** at low temperatures we may not trust the measured transition temperatures

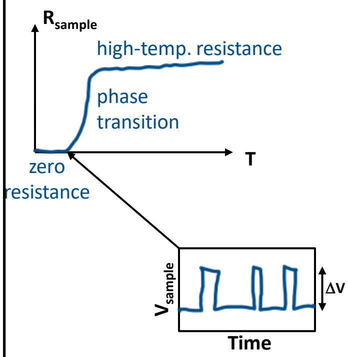
(iii) The transition is different when the sample is cooled (blue) or warmed (red), i.e. a hysteresis is observed. Reason: the thermometer and the sample are not in thermal equilibrion, e.g. the themrometer cools and warms faster than the sample.
SOLUTION: well-controlled slow cooling and warming with a **PID temperature controller** (see later).



Measuring the zero resistance of a high-temperature superconductor sample

How accurate is the measurement?

20mm x 3mm x 2mm



Resolution of the four-probe resistance measurement according to the digital voltage measurement error:

Measuring current: I_{sample}=50mA

Digital resolution of the voltmeter: ΔV=0.01mV

Resolution of the resistance measurement: ΔR= 0.2mΩ

Dimensions: h=20mm, A=2mm x 3 mm

-> Resistivity resolution: $\Delta\rho = \Delta R\ A/h = 6 \times 10^{-8}\ \Omega m$

As a comparison, the resistivity of copper at room temperature: $\rho_{Cu,T=300K} = 1.7 \times 10^{-8}\ \Omega m$!!!

We do see a sudden transition, but the error of measuring zero resistance is larger than the resistance of a similarly sized copper rod would be.

Can we use a completely different measurement method to reduce the resistance measurement error by several orders of magnitude?

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Measuring the zero resistance of a high-temperature superconductor sample – persistent current measurement

Measurement trick: measuring circulating superconducting current by a magnetometer

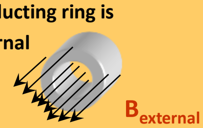
If we could put **current** to a **superconducting ring**, it would **circulate forever** due to the zero resistance. This persistent current is easily **measured by a magnetic field sensor**.

How can we put this current to the ring? Let us cool down the superconductor in external magnetic field. Inside the ring the Φ=BA flux changes in time as:

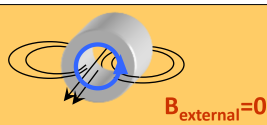
$$-\frac{\mathrm{d}\Phi}{dt} = V = RI = 0$$

Where R is the resistance of the ring, A is its cross section, and B the magnetic field inside. In case of zero resistance the flux cannot change inside, i.e. once the external field is switched off, a circulating current should arise in the ring, which maintains the original magnetic field inside the ring. This way it is easy to establish a circulating persistent current.

The superconducting ring is cooled in external magnetic field



Once the field is switched off a circulating current starts which maintains the original field



How fast would the current decay in an a ring with finite resistance?
(L is the ring's self-inductance and τ is the characteristic decay-time)

$$RI = -\frac{\mathrm{d}\Phi}{dt} = -L\frac{\mathrm{d}I}{\mathrm{d}t} \rightarrow B(t) \sim I(t) = I_0 e^{-\overbrace{(R/L)}^{1/\tau} t}$$

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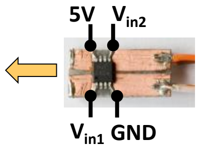
Measuring the zero resistance of a high-temperature superconductor sample – persistent current measurement

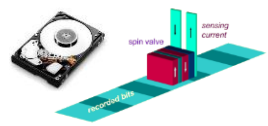


GMR magnetic field sensor

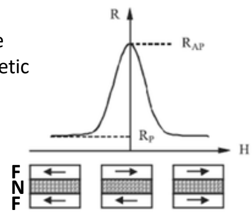
A. Fert és P. Grünberg

Nobel prize: 2007.

Applied sensor: The persistent current of the ring can be measured by a Giant Magneto Resistance (GMR) sensor.



GMR (Giant MagnetoResistance) phenomenon: electrons pass through such a nanostructure which consists of a nonmagnetic (N) layer between two ferromagnetic (F) layers. The resistance is sensitive for the relative orientation of magnetization in the two F layers.



This phenomenon is used in the reading head of spinning hard disk drives. The magnetization direction of one F layer is fixed, while the direction of the other is set by the magnetization of the disk (this is called spin valve).

Explanation: The magnetic layers polarize the spin of electrons passing through. The arrangement is similar to optical polarizers: light can only pass through two polarizers of the same orientation, while it cannot pass through them if the orientations are perpendicular. The GMR phenomenon occurs only in nanostructures if the two magnetic layers are very close to each other.

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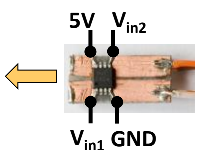
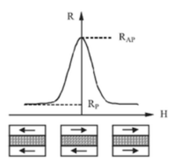
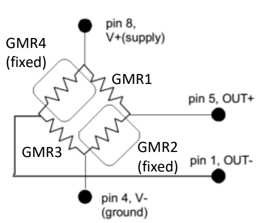
Measuring the zero resistance of a high-temperature superconductor sample – persistent current measurement



GMR magnetic field sensor



Measurement trick: bridge arrangement



Instead of a single sensor, 4 similar GMR sensors are applied in a bridge arrangement. In two of these (GMR2 and GMR4) the antiparallel magnetic orientation of the layers is fixed, it cannot be changed by the external field.

Voltage is applied at two opposite corners of the bridge (pins 4 & 8) and voltage is measured at the other two corners (pins 5 & 1).

If the field is zero, but the temperature changes, the resistance of the 4 sensors changes in the same fashion, i.e. the bridge remains balanced, and the output voltage difference remains zero.

If a magnetic field is applied, the fixed GMR sensors (GMR2 and GMR4) keep their resistance, while the resistance drops for the working GMR sensors (GMR1 and GMR3), and thereby the unbalanced bridge outputs a finite voltage, which is proportional to the strength of the magnetic field.

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Measuring the zero resistance of a high-temperature superconductor sample resistance vs. persistent current measurement

We have seen that our four probe method measures the zero resistivity with $\Delta\rho = 6 \times 10^{-8}\ \Omega\text{m}$ error due to the finite digital resolution. What would be the τ current decay time constant in our superconducting ring, if its resistivity was not zero, rather the above $\Delta\rho = 6 \times 10^{-8}\ \Omega\text{m}$ resistivity error value?

Height: h=15mm
Diameter: d=15mm
Wall thickness: w=1.5mm

$\tau = L/R$, $R = d\pi\,\rho/(h\,w) = 0.125\,\text{m}\Omega$, $L = \mu_0 A/h = 4\pi \times 10^{-7} \times d^2\pi/(4h) = 1.5 \cdot 10^{-8}$ H -> $\boldsymbol{\tau = 120\,\mu s}$ **than the four-probe resistance measurement.**

In case of persistent current measurement, even if the magnetic field measurement is somewhat unstable, we may say, that the current decays by less than 10% within a time of t=10minutes=600s.
-> $I(t) = I_0 \exp(-t/\tau) > 0.9 \cdot I_0$ -> $\boldsymbol{\tau > 6000\text{s}}$

This calculation shows, that **the persistent current measurement determines the zero resistance with almost 8 orders of magnitude better resolution than the four-probe resistance measurement.**

The magnetic field of a persistent solenoid magnet in a Nuclear Magnetic Resonance (NMR) spectrometer was measured for several years. From this the current decay time-constant of the superconductor turned out to be $\tau > 10^5$ years, with the corresponding resistivity $\rho_{sc} < 10^{-32}\ \Omega\text{m}$!!
J. File & R. G. Mills, Phys. Rev. Lett 10, 93 (1963)

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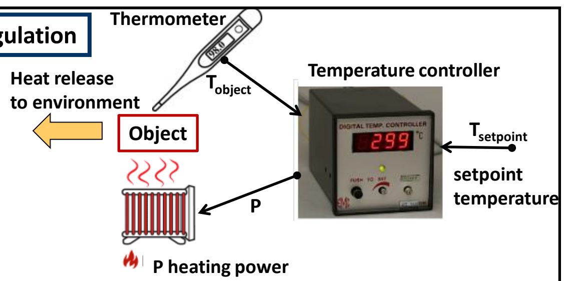
Precise temperature control: PID regulation



A simple control task: an object is to be heated to a given $T_{setpoint}$ target (setpoint) temperature by a heating element. The temperature (T) is measured and the heating power (P) is controlled based on the difference between the measured temperature and the setpoint temperature.

Simplest implementation: the operation of a room thermostat: $P = \begin{cases} P_0, \text{if } T_{\text{object}} < T_{\text{setpoint}} \\ 0, \text{if } T_{\text{object}} > T_{\text{setpoint}} \end{cases}$

More elaborate, **proportional regulation:** $P = \begin{cases} C_p \cdot (T_{\text{setpoint}} - T_{\text{object}}), \text{if } T_{\text{object}} < T_{\text{setpoint}} \\ 0, \text{if } T_{\text{object}} > T_{\text{setpoint}} \end{cases}$

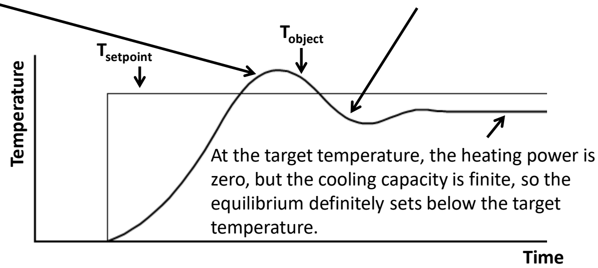
If the object cools below the target temperature, the heating is switched on in proportion to the temperature difference, but the system does not react fast enough and cools below the target temperature.

Example: the target temperature is raised, how the control reacts:

As the object heats up above the target temperature, the heating is turned off, but the object continues to heat up due to the heat inertia.

If the C_p is too high, the overheating-overcooling oscillation becomes stationary, and no equilibrium is reached.

If the C_p value is too low, we cannot reach the target temperature at all.



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PID control

More precise control: the integral of the error signal is also taken into account **(so-called PI control):**

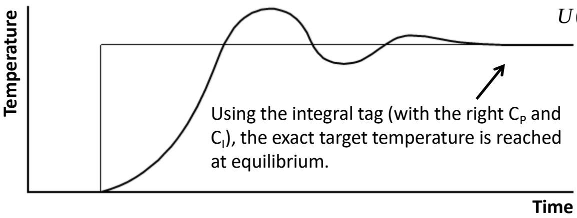
$$\text{output}(t) = C_P\big(T_{\text{setpoint}} - T_{\text{object}}(t)\big) + C_I \int_0^t \big(T_{\text{setpoint}} - T_{\text{object}}(\tau)\big)\,\mathrm{d}\tau\,, \qquad P = \begin{cases} \text{output, if output} > 0 \\ 0\text{, if output} < 0 \end{cases}$$

Proportional constant

Integral constant

This is similar to the equation for a series RC circuit:

$$U(t) = R \cdot I(t) + \frac{1}{C}\int_0^t I(\tau)\,\mathrm{d}\tau$$



The voltage of a series RC circuit can change over a characteristic time τ_{RC}=RC. Based on this analogy, the PI control "responds" with a characteristic time of $\tau_{PI}=C_P/C_I$.

The τ_{PI} time constant of the PI controller should be taken to be approximately the same as the time constant of the system due to thermal inertia (see period of oscillation due to thermal inertia)

Note, that in equilibrium ($T_{\text{setpoint}}=T_{\text{object}}$) the integral term is a time-independent (constant) heating power, which actually balances the cooling power of the environment.

The control can be supplemented with a differential term (so-called **PID control**):

$$\text{output}(t) = C_P\big(T_{\text{setpoint}}(t) - T_{\text{object}}(t)\big) + C_I \int_0^t \big(T_{\text{setpoint}}(\tau) - T_{\text{object}}(\tau)\big)\mathrm{d}\tau + C_D \frac{\mathrm{d}\big(T_{\text{setpoint}}(t) - T_{\text{object}}(t)\big)}{\mathrm{d}t}$$

Proportional constant

Integral constant

Differential constant

This is useful if the target value also changes over time. In the case of a sudden target value change, the differential term makes the PID control respond faster than if it were only PI control.

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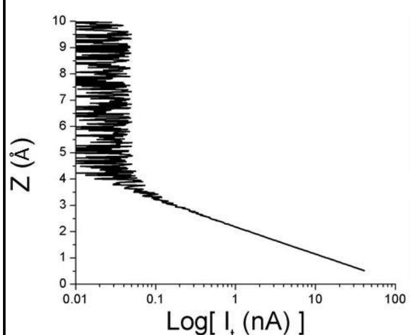
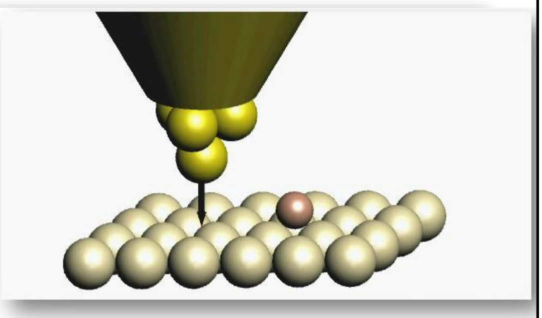
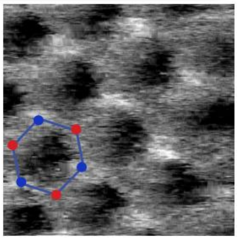
Another example of PI control: scanning tunnelling microscope (STM)





Gerd Binnig, Heinrich Rohrer
Nobel prize 1986.

A quantum mechanical tunnel current flows between a metal tip and a nearby metal surface even when the tip is not touching the surface. The tunnelling current is an exponential function of the tip-sample distance, if the tip is lifted by half an atom-atom distance (~0.1 nm) the tunnelling current decreases by an order of magnitude.



STM image of graphite

„For their design of the scanning tunneling microscope"

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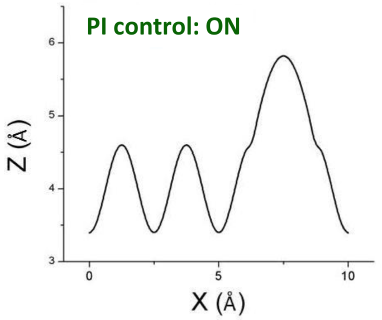
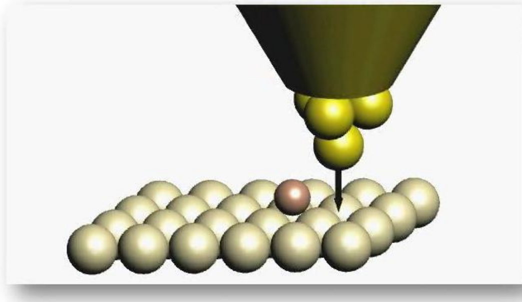
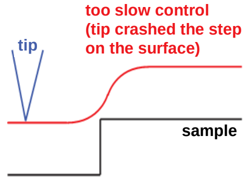
Another example of PI control: scanning tunnelling microscope (STM)

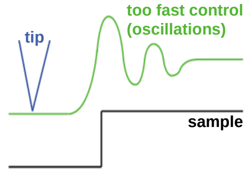




As we sweep the tip over the surface, we raise or lower the tip in the direction perpendicular (z) to the surface so that the tunnel current, i.e. the tip-sample distance, remains the same. This can be achieved using a PI control circuit:

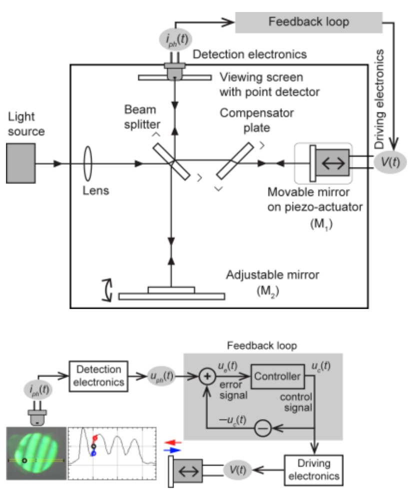
$$\frac{dz}{dt} = C_P\left(I(t) - I_{\text{setpoint}}\right) + C_I \int_0^t \left(I(t) - I_{\text{setpoint}}\right)d\tau$$





Here the PI control is not giving the elevation, but the rate of change of elevation (tip velocity).

Yet another example of PI control: Michelson interferometer



What, if we want to measure larger than wavelength displacements with sub-wavelength resolution?

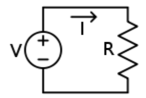
In normal operation, **it is hard to do that, we should count the oscillations** also knowing which direction the object moves.

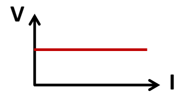
Solution: PI control. The object under study is moving at one mirror. At the other mirror, we move the piezo such that its displacement completely follows the displacement of the studied object. If we succeed, the intensity at the viewing screen will be constant in time.

To do that we actually choose a setpoint intensity halfway between constructive and destructive interference, **we measure the intensity with a photodetector, and we control the piezo actuator such, that the measured intensity is best stabilized at the setpoint intensity**. Like this we can read the displacement of the studied object from the displacement of the feedback-controlled piezo actuator spanning very large displacements with sub-wavelength resolution.

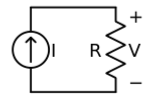
Voltage and current sources, voltage and current meters

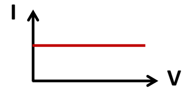
Ideal voltage generator: a voltage source whose output voltage does not depend on the resistance (or current) of the load circuit



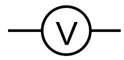


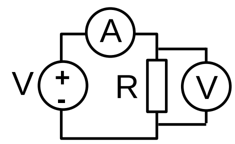
Ideal current generator: a current source whose current does not depend on the resistance (or voltage) of the load circuit



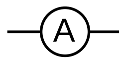


Ideal voltmeter: voltmeter with infinite internal resistance, connected in parallel with the circuit element to be measured. The presence of the voltmeter does not affect the current flowing through the measured object!





Ideal current meter: a current meter with zero internal resistance, connected in series with the circuit element to be measured. The presence of the current meter does not affect the voltage on the measured object!



On most power supplies both voltage and current can be set. From the both, the more restricting one is realized. Example: 30V & 10A is set at 100Ω load. Current of 10A would require 1000V, i.e. 30V is realized on the output at 0.3A current, and the CV light is on (constant voltage). Example 2: at 1Ω resistor 10A and 10V is realized (CC-constant current light is on).

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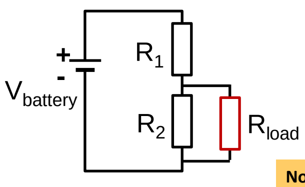
Simple voltage and current generator circuit

A battery and a voltage divider as a voltage source:

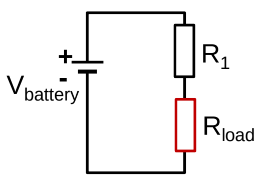


$$V_{\text{load}} = V_{\text{battery}} \cdot \frac{R_2}{R_1 + R_2}, \text{if } R_{\text{load}} >> \mathrm{R}_2 = \mathrm{R}_{\text{output}}$$

Output resistance of the voltage source (ideally zero). The desired voltage is only obtained, if the load resistance is much larger than the output resistance.

Notes: such battery circuits are not only simplified examples, but a battery is also an ideal choice if a source with very low internal noise level is desired. The disadvantage is that the voltage of the battery is not adjustable. As a further note, the above voltage divider circuit is also useful, if an adjustable voltage source is used instead of a battery, as the voltage division also divides the output noise level (i.e. for 1mV output it might be better to output 100 mV with 1:100 division, as the direct 1mV setting of the voltage source may have larger output noise level).

A battery and a resistor as a current source:



Output resistance of the current source (ideally infinity). The desired current is only obtained if the load resistance is much smaller than the output resistance.

$$I_{\text{load}} = \frac{V_{\text{battery}}}{R_1}, \text{if } R_{\text{load}} \ll \mathrm{R}_1 = \mathrm{R}_{\text{output}}$$

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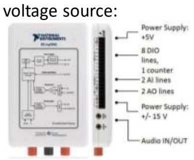
A data acquisition (DAQ) card is a general-purpose instrument which acts both as a versatile voltmeter and a programmable voltage source:



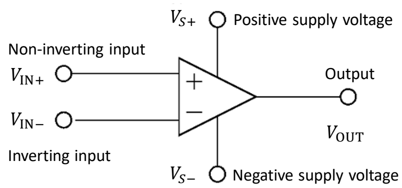
Inputting and outputting arbitrary voltage patterns

The operation of general digital voltage sources and voltmeters relies on analog-to-digital (A/D) and digital-to-analog (D/A) converters. These devices include several amplifiers. Though the operation of amplifiers is a subject of electronics courses, here a brief insight is given to A/D and D/A converters as well as the basics of amplifier circuits.

Basics of operational amplifiers (op-amps) treated as a black-box:
The op-amp is an integrated circuit with multiple pins:

- One positive and one negative supply voltage (e.g. +15V and -15V): V_{S+}, V_{S-}
- Two input voltages (a non-inverting and an inverting one): V_{IN+}, V_{IN-}
- An output voltage: V_{OUT}

The **amplified input voltage difference is outputted**: $V_{OUT} = A_{OL} \cdot (V_{IN+} - V_{IN-})$, where A_{OL} is the so-called **open loop voltage gain**. A_{OL} is usually very large (e.g. $A_{OL} \approx 10^5$). If the output voltage would approach the positive/negative supply voltage, rather a so-called saturation voltage is outputted (V_{sat+}, V_{sat-}), which is close to the positive/negative supply voltage, see red curve.
Practically **no current flows at the two inputs** (V_{IN+}, V_{IN-}), i.e the input resistance is practically infinite.



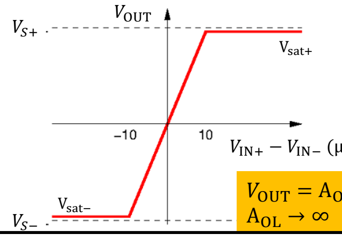


Figure source: Varga-Bagoly: Electronics and measurement techniques (in Hungarian)

$$V_{OUT} = A_{OL} \cdot (V_{IN+} - V_{IN-})$$
$$A_{OL} \to \infty$$

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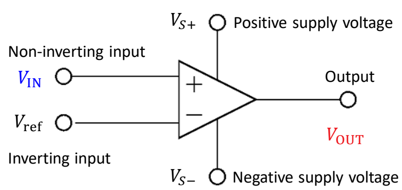
Using an op-amp as a comparator

Simplest application: comparator circuit. A reference voltage V_{ref} is connected to the negative input, and the investigated input voltage V_{IN} is connected to the positive input. Due to the close-to-infinity (very large) open loop gain already a relatively small voltage difference compared to V_{ref} yields a saturated voltage at the output such that:

- $V_{OUT}=V_{sat-}$, if $V_{IN} < V_{ref}$,
- $V_{OUT}=V_{sat+}$, if $V_{IN} > V_{ref}$.

Therefore, the output voltage can be used to decide whether the V_{IN} input signal is greater or less than the V_{ref} reference signal (the input is *compared* to the reference -> comparator).



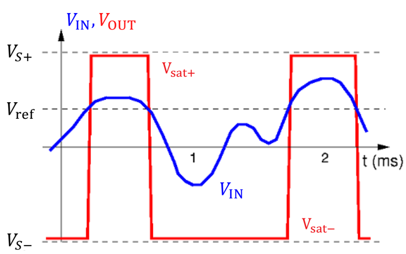


Figure source: Varga-Bagoly: Electronics and measurement techniques (in Hungarian)

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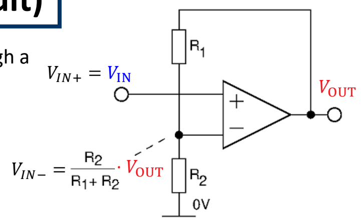
Using the op-amp as a voltage amplifier (non-inverting voltage amplifier circuit)

Figure source: Varga-Bagoly: Electronics and measurement techniques (in Hungarian)



The output of the amplifier is fed back to the negative input through a voltage divider with R_1 and R_2 resistances (***negative feedback***):

$V_{\mathrm{IN}-} = \frac{R_2}{R_1+R_2} V_{\mathrm{OUT}}$, furthermore, $V_{\mathrm{OUT}} = A_{\mathrm{OL}} \cdot (V_{\mathrm{IN}+} - V_{\mathrm{IN}-})$

From these: $\frac{R_1+R_2}{R_2} V_{\mathrm{IN}-} = A_{\mathrm{OL}} \cdot (V_{\mathrm{IN}+} - V_{\mathrm{IN}-})$

From these, also considering the $A_{\mathrm{OL}} \to \infty$ condition:

$V_{\mathrm{IN}-} = V_{\mathrm{IN}} \cdot \frac{R_2 A_{\mathrm{OL}}}{R_2 A_{\mathrm{OL}} + R_1 + R_2} \approx V_{\mathrm{IN}}$, i.e. due to the negative feedback the potentials of the „+" és „-" inputs are the same: $V_{\mathrm{IN}} = V_{\mathrm{IN}+} = V_{\mathrm{IN}-}$.

From all these

$$V_{\mathrm{OUT}} = \frac{R_1 + R_2}{R_2} V_{IN}$$

follows, i.e. the proper voltage amplification can be set by the R_1 and R_2 resistances

General rules for calculating amplifier circuits:
(i) The current is zero at the two inputs (i.e. both the inverting and non-inverting input)
(ii) The inverting and non-inverting inputs have the same potential ($V_{\mathrm{IN}+} = V_{\mathrm{IN}-}$)

Note, that rule (ii) is not general for all amplifier circuits, it only applies for cases with negative feedback (i.e. the output is fed back to the negative input). I.e. rule (ii) does not apply for a comparator circuit, but it applies to most voltage and current amplifier circuits, as all these rely on negative feedback.

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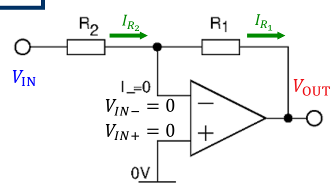
Using the op-amp as a voltage amplifier (inverting voltage amplifier circuit)

Figure source: Varga-Bagoly: Electronics and measurement techniques (in Hungarian)



The positive input is grounded ($V_{IN+} = 0$) and the output is fed back to the negative input through the R_1 resistor. Due to rule (ii) the negative input is also at virtual ground, $V_{IN-} = 0$. The current on the R_2 resistor is calculated as:

$I_{R_2} = \frac{V_{\mathrm{IN}} - V_{\mathrm{IN}-}}{R_2} = \frac{V_{\mathrm{IN}}}{R_2}$

Furthermore, the I_{R_2} current flowing through R_2 cannot flow into the negative input of the amplifier (see rule (i)), i.e. the entire I_{R_2} current flows through the R_1 resistor as well ($I_{R_2} = I_{R_1}$):

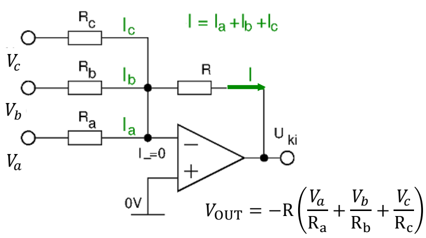
$I_{R_1} = \frac{V_{\mathrm{IN}-} - V_{\mathrm{OUT}}}{R_1} = -\frac{V_{\mathrm{OUT}}}{R_1} = I_{R_2} = \frac{V_{\mathrm{IN}}}{R_2}$

From the $I_{R_2} = I_{R_1}$ condition

$$V_{\mathrm{OUT}} = -\frac{R_1}{R_2} V_{IN}$$

follows, i.e. the proper voltage amplification can be set by the R_1 and R_2 resistance ratio.

Generalization: summing amplifier



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Digitally adjustable voltage source: D/A converter

A D/A converter (DAC, digital-to-analog converter): converts a digital signal sequence into an analog voltage.

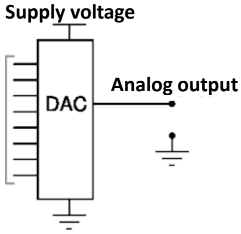
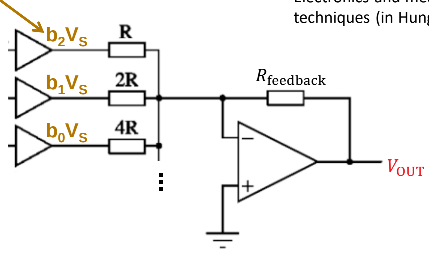


Figure sources: Varga-Bagoly: Electronics and measurement techniques (in Hungarian)

One of the simplest implementations:
binary-weighted DAC (3 bits in the figure):

Binary numbers (b_0, b_1, b_2) are fed to the inputs. In practice the 0 or 1 binary input is represented by 0V or V_S~5V voltage at the input channels. The resistances of the channels increase according to the powers of 2.



According to the operation of the summing amplifier the output voltage is:

$$V_{OUT} = -\frac{V_S R_{feedback}}{2^N R}\sum_{i=0}^{N} 2^i b_i\,, \qquad b_i = 0{,}1, V_S \approx 5V$$

i.e. the output is effectively a conversion of the input binary number to decimal number. The latter is scaled by the values of R, $R_{feedback}$ and V_S to the desired output voltage range.

Example with 3 bits:
The input is a number encoded in binary (e.g. 101). The summing circuit uses increasing resistors in powers of 2 to convert the binary input signal to an analog value. Example: if the gains are adjusted to give an output of V_{OUT}=1V for the least significant bit (001), then the output in volts is simply the decimal value of the input binary signal, e.g., an input of (101) gives an output of 5V. This device requires a single op-amp and N binary inputs for N bit resolution.

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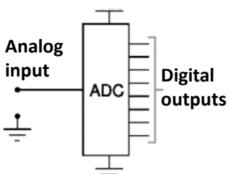
Digital voltmeter: A/D converter

A/D converter (ADC, analog-to-digital converter): converts an analog voltage into a digital (binary) signal. A small signal is first amplified and afterwards A/D conversion is applied.

Flash ADC: Equidistant voltage values are generated from a reference voltage V_{ref} using a resistor ladder, and these voltages are connected to the negative inputs of the A/D converter's comparator amplifiers (e.g. for V_{ref}=7V, the negative inputs are at 1,2, ..., 7 V, as shown in the figure below). Each comparator compares the voltage V_{in} applied to the input of the A/D converter with the reference voltage from the resistor ladder (e.g. for V_{in}=5.2V, the output of the bottom five amplifiers is V_{sat+} ("1" digital value) and the output of the other amplifiers is V_{sat-} ("0" digital value), i.e. a sequence of 1111100 numbers is obtained, which is digitally converted to a binary number system, i.e. the number of "1" values (5 in this case) is converted to a binary number (->101).



The parallel (flash) ADC is perhaps the simplest in principle, the fastest, and also the most expensive (requiring the most components) converter. A 3-bit parallel ADC circuit is shown in the figure. Note, that for N bit resolution 2^N-1 comparators are required (7 comparators for 3bit resolution.)

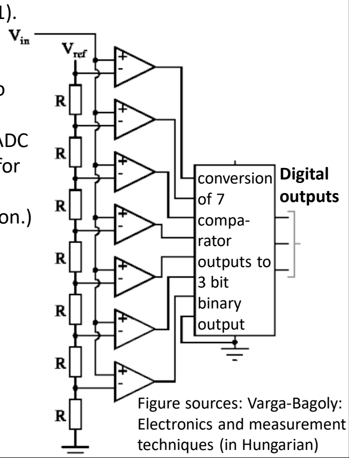


Figure sources: Varga-Bagoly: Electronics and measurement techniques (in Hungarian)

Sucessive approximation ADC: a single comparator amplifier is used, whose reference voltage is generated by a D/A converter. The output voltage of the DA converter is varied by a bisection method to find the digital voltage value closest to the input voltage based on the result of the comparator. (In the example first it is checked, if the input is larger or smaller than 4V (middle of the 0..8V range). If larger, the 4..8 V range is halved, i.e. the input is compared to 6 V. If smaller than 6V, the 4..6V range is halved, i.e. the input is compared to 5V.)

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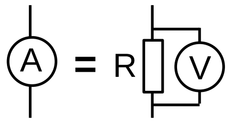
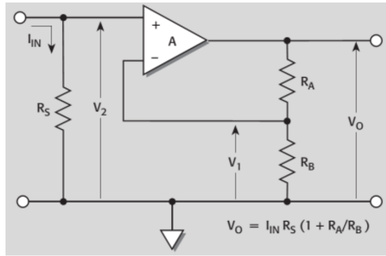
Current measurement

Source: Keithley Low Level Measurements Handbook

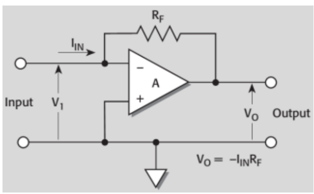
Shunt current meter circuit:

Simplest method: measure the voltage across a small resistor with a voltage amplifier.





Current amplifier circuit:



Similar to the inverting voltage amplifier circuit, just the R_2 resistor at the input is missing. The positive input is grounded, and therefore the negative input (i.e. the current input of the amplifier) is also at virtual ground (see rule (ii) of op-amps). According to rule (i) the entire input current flows towards the output, i.e. the output voltage is:

$$\mathrm{V_{OUT}} = \underbrace{V_{IN-}}_{0} - I_{\mathrm{IN}} \cdot R_{\mathrm{F}} = -R_{\mathrm{F}} \cdot I_{IN}$$

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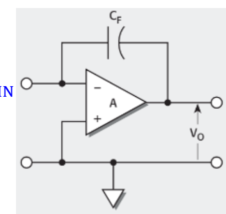
Charge amplifier

Source: Keithley Low Level Measurements Handbook

Q_{IN}

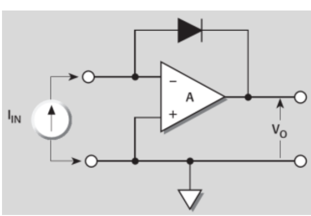


$$\mathrm{V_{OUT}} = -\frac{Q_{\mathrm{IN}}}{C_{\mathrm{F}}}$$

Again, the positive input is grounded, and therefore the negative input (i.e. the charge input of the amplifier) is also at virtual ground (see rule (ii) of op-amps). According to rule (i) no current can flow at DC operation, and the Q=CV relation for a capacitor yields the demonstrated voltage at the output.

Note: in case of AC operation the capacitor can be considered as a circuit element with Z=1/(iωC) complex impedance from which $\mathrm{V_{OUT}} = -I_{IN}/(i\omega C_{\mathrm{F}})$ follows. This is practically a current amplifier with frequency dependent amplification and a 90-degree phase shift.

Logarithmic current amplifier



$$\mathrm{V_{OUT}} \sim \log(I_{\mathrm{IN}})$$

In the current amplifier circuit a diode is used instead of a feedback resistor, for which a I~exp(const. * V) exponential current-voltage characteristic is observed. From this the output voltage is practically the logarithm of the input current. This amplifier is useful, if a very broad current range (many orders of magnitude) should be covered.

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Device specifications

Note the difference between accuracy and resolution: a ruler with a resolution of 1 mm does not necessarily have an accuracy of 1 mm.

In the figure the scale of the wooden ruler is misprinted, yielding 3mm error at 30cm reading. From this the accuracy is 1mm+1% of the reading.

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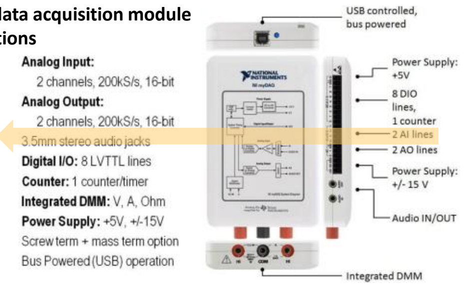
Analog inputs of a data acquisition card (DAQ)

National Instruments myDAQ data acquisition module specifications



Analog inputs (see 2AI lines)
ADC** (Analog digital converter) **resolution: 16 bit.
Voltages in the range of +-10 V are measured. This analog input voltage is converted to a digital signal, i.e. a 16 bit binary number. This means that the 20V input voltage range is divided to 2^{16}= 65536 resolvable segments, i.e. the **resolution** for a 16 bit card and a +-10V input range is 20/65536 V= **305µV**

Possible input ranges: +-10V, +-2V - in the +-2V input range first an 5x amplification is applied, and afterwards the ADC with +-10V input range digitizes the input signal.

Absolute accuracy – the error of the voltage measurements on the AI lines compared to the true values:

Nominal Range		Typical at 23 °C (mV)	Maximum (18 to 28 °C) (mV)
Positive Full Scale	Negative Full Scale		
10	–10	22.8	38.9
2	–2	4.9	8.6

accuracy *resolution* 0.3mV

Note the two orders of magnitude difference between the accuracy and the resolution! Still we can detect 0.3mV changes!

Maximum sampling rate: 200kS/s –maximum 200000 voltage values are sampled in one second

Input impedance >10GΩ – very high internal resistance

Overvoltage protection +-16V - at higher voltage the input is damaged.
The absolute maximum rating is the first to read in the device manual!

Common mode rejection ratio (CMRR) (DC to 60Hz): 70dB (see later)

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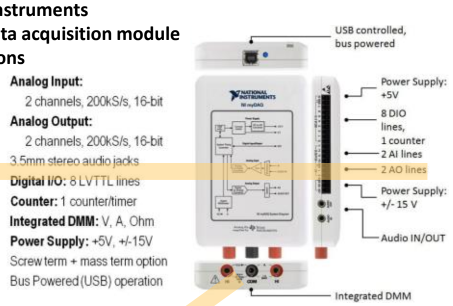
Analog outputs of a data acquisition card (DAQ)

National Instruments myDAQ data acquisition module specifications



Analog outputs (see 2AO lines)
DAC** (Digital to analog converter) **resolution: 16 bit.
Similarly, the +-10 V output range is divided to $20V/2^{16}$= 305µV steps.

Further specifications:
Range: +-10V, +-2V;
Maximum update rate: 200kS/s;
Output impedance: 1Ω
Maximum output current: 2mA – e.g. a 1V output cannot supply 1V/ 10Ω=100mA to a 10Ω load, even if this load is larger than the output impedance. Instead, it can only output 2mA.

Digital multimeter input (DC voltage measurement)

Resolution: 3.5 digit
full digit: 0-9; half digit: 0-1,
i.e. a 3 digit device displays in the 0-999-range,
whereas a 3.5 digit in the 0-1999 range

Function	Range	Resolution	Accuracy ± ([% of Reading] + Offset)
DC Volts	200.0 mV	0.1 mV	0.5% + 0.2 mV

3.5 digit: the 200mV range is divided to 2000 parts

If we measure 100mV in 200mV range: the accuracy is 100mV*0.5% + 0.2mV = 0.7mV (> 0.1mV resolution!!)

illustration of 3.5 digits

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Conversion of possible resolution specifications

PPM: part per million

Decibel (dB): $20 \cdot \log_{10}\left(\frac{V_{res}}{V_{range}}\right)$

Example The V_{res} voltage resolution is 1% of the full voltage range (V_{range})

Percent	PPM	Digits	Bits	dB	Portion of 10V
10%	100000	1	3.3	−20	1 V
1%	10000	2	6.6	−40	100mV
0.1%	1000	3	10	−60	10mV
0.01%	100	4	13.3	−80	1mV
0.001%	10	5	16.6	−100	100 µV
0.0001%	1	6	19.9	−120	10 µV
0.00001%	0.1	7	23.3	−140	1 µV
0.000001%	0.01	8	26.6	−160	100 nV
0.000001%	0.001	9	29.9	−180	10 nV

Further possible specification using the datasheet of a Keithley 2001 7.5 digit DMM
-Temperature coefficient: +-(% of reading + % of range)/ °C
-Accuracy on different time scales (this means that accuracy degrades over time, i.e. it is not a bad idea to recalibrate a sensitive instrument at certain intervals!).

DCV INPUT CHARACTERISTICS AND ACCURACY

RANGE	FULL SCALE	RESO-LUTION	DEFAULT RESO-LUTION	INPUT RESISTANCE	ACCURACY[4] ±(ppm of reading + ppm of range) 5 Minutes[5]	24 Hours[6]	90 Days[7]	1 Year[7]	2 Years[7]	TEMPERATURE COEFFICIENT ±(ppm of reading + ppm of range)/°C Outside T_{CAL} ±5°C
200mV[8]	±210.0mV	10nV	100nV	>10GΩ	3 + 3	10 + 6	25 + 6	37 + 6	50 + 6	3.3 +1.5
2V	±2.10V	100nV	1µV	>10GΩ	2 + 1.5	7 + 2	18 + 2	25 + 2	32 + 2	2.6 +0.15
20V	±21.0V	1µV	10µV	>10GΩ	2 + 1.5	7 + 4	18 + 4	24 + 4	32 + 4	2.6 +0.7
200V	±210.0V	10µV	100µV	10MΩ±1%	2 + 1.5	13 + 3	27 + 3	38 + 3	52 + 3	4.3 +1
1000V	±1100.0V	100µV	1mV	10MΩ±1%	10 + 1.5	17 + 6	31 + 6	41 + 6	55 + 6	4.1 +1

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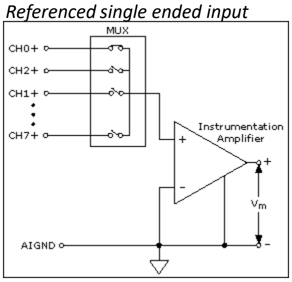
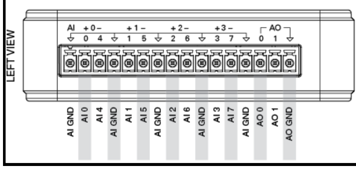
Possible DAQ card input types

It's important to note that most DAQ boards multiplex the inputs, i.e. one amplifier and one A/D converter measures all the inputs, with just switches to toggle which one is being measured. This also means that with a 200kS/s card, for example, a maximum sampling rate of 100kS/s per channel can be achieved by measuring 2 channels.

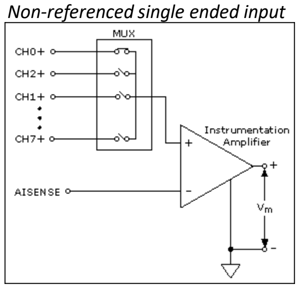
Referenced single ended input



Negative points for all channels are connected to AIGND, which is on the mains ground (e.g. we measure between AI0 and AIGND)

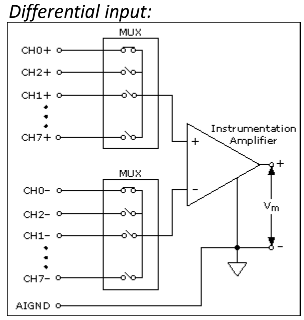


Non-referenced single ended input



Negative points for all channels are connected to AIGND, which is independent of the mains ground. (e.g. we measure between AI0 and AIGND, but in this setting the card internally disconnects AIGND with a switch from the mains ground)

Differential input:



True differential measurement, the card measures the voltage difference between two inputs (e.g. AI0 and AI4, or AI1 and AI5). This halves the number of input channels!

Common mode rejection

Common mode rejection ratio (CMRR), *e.g. for the analog input of a MyDAQ card: CMRR (DC to 60Hz): 70dB*

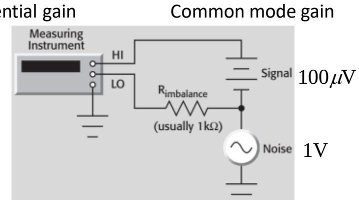
Ideally, the output voltage of an operational amplifier depends only on the difference in input voltages. In the real case, however, if both inputs are around the same non-zero voltage, called common mode, a finite output voltage will appear. With a simple model:

$$V_{out} = \underbrace{A_D}_{\text{Differential gain}} \cdot (V_+ - V_-) + \frac{1}{2}\underbrace{A_{CM}}_{\text{Common mode gain}}(V_+ + V_-)$$

The suppression of the common mode signal is characterized as:

$$CMRR = \left|\frac{A_D}{A_{CM}}\right|; \quad CMRR\,[dB] = 20\log_{10}\left(\left|\frac{A_D}{A_{CM}}\right|\right)$$

Conceptually, CMRR tells you how accurately small voltage differences can be measured far away from zero voltage, in the example figure the question is, how accurately a small (100µV) signal can be measured above a large background voltage (1V). CMRR may depend on frequency, so it is often given as a function of frequency.



Example: to measure a voltage difference of 100µV between HI and LO inputs with a DC background of around 1V, what is the measurement error at 70dB CMRR?
The instrument amplifies the measured signal before A/D conversion, according to the above model this amplified signal is: $V_{out}=A_D$*100µV+A_{CM}*1V. From this the measured signal is deduced as $V_{measured}=V_{out}/A_D$, i.e the measurement error of the 100µV signal due to the common mode amplification is **ERROR =1V*A_{CM}/A_D=1V/CMRR**

CMRR[dB]=70dB → CMRR=$10^{3.5}$

ERROR=1V/ $10^{3.5}$=316µV, i.e. the error is larger than the signal to be measured. Without the 1V DC background signal, i.e. if the 100µV would be measured around zero voltage backgroung, this error would not appear.

As a reference a **Keithley 2001 DMM with CMRR=140dB would yield** a much smaller error: **ERROR=1V/ 10^7=0.1µV!**

High speed measurements

So far the speed (or frequency) of the measurement was not an issue in the discussions. Next, we treat various aspects of high frequency measurements, such as:

- Synchronization of signals, triggering strategies
- Special techniques for ultrafast sampling
- Aliasing
- RC time constant limitations
- Wave propagation in coaxial cables, transmission and reflection on circuit elements.

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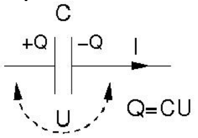
Reminder: complex impedance

Figure sources: Varga-Bagoly: Electronics and measurement techniques (in Hungarian)

Capacitor:



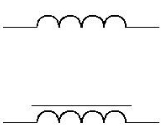
$$Q = CU$$

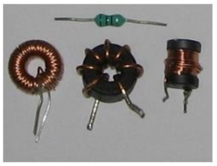
$$I = C\frac{\mathrm{d}U}{\mathrm{d}t}$$

$$U_\omega(t) = \hat{U}_0 e^{i\omega t} \;\Rightarrow\; I_\omega(t) = C\cdot i\omega \hat{U}_0 e^{i\omega t} = i\omega C\cdot U_\omega(t)$$

$$\Rightarrow U_\omega(t) = Z_C(\omega) I_\omega(t),\quad Z_C(\omega) = \frac{1}{i\omega C}$$

Inductor:





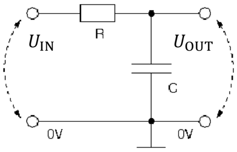
$$U = L\frac{\mathrm{d}I}{\mathrm{d}t}$$

$$I_\omega(t) = \hat{I}_0 e^{i\omega t} \;\Rightarrow\; U_\omega(t) = L\cdot i\omega \hat{I}_0 e^{i\omega t} = i\omega L\cdot I_\omega(t)$$

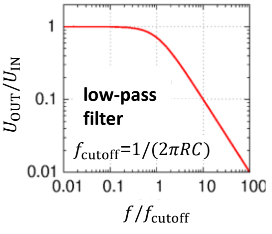
$$\Rightarrow U_\omega(t) = Z_L(\omega) I_\omega(t),\quad Z_L(\omega) = i\omega L$$

Example: RC filter

input



output



$$\frac{U_{OUT}}{U_{IN}} = \frac{Z_C}{Z_R+Z_C} = \frac{\frac{1}{i\omega C}}{R+\frac{1}{i\omega C}} = \frac{1}{1+i\omega RC}\cdot\frac{1-i\omega RC}{1-i\omega RC} = \frac{1-i\omega RC}{1+(\omega RC)^2} \;\Rightarrow\; \left|\frac{U_{OUT}}{U_{IN}}\right| = \frac{\sqrt{1+(\omega RC)^2}}{1+(\omega RC)^2} = \frac{1}{\sqrt{1+(\omega RC)^2}}$$

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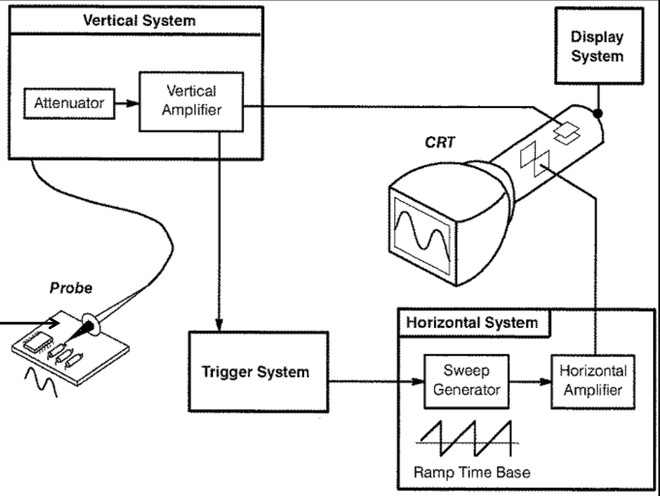
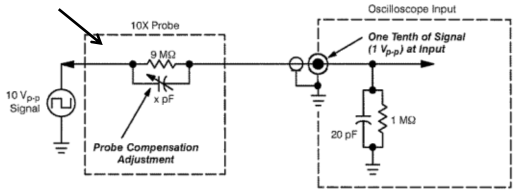
Analog oscilloscope



Input signal is divided as needed (attenuator), or amplified (vertical amplifier). The resulting signal will control the vertical displacement of the cathode ray.

Optionally, we connect special measuring probe to the input

Most probes apply 1:10 voltage division,and at the same time, increasing the input resistance by a factor of 10, so it has less influence on the measured circuit.



The input signal is watched by a synchronizing (trigger) circuit: when the signal reaches a certain threshold (trigger level) with the desired slope (trigger slope), the synchronizing circuit releases a sawtooth signal, that controls the horizontal displacement of the cathod ray. The set time base on the oscilloscope corresponds to the period of the sawtooth signal.

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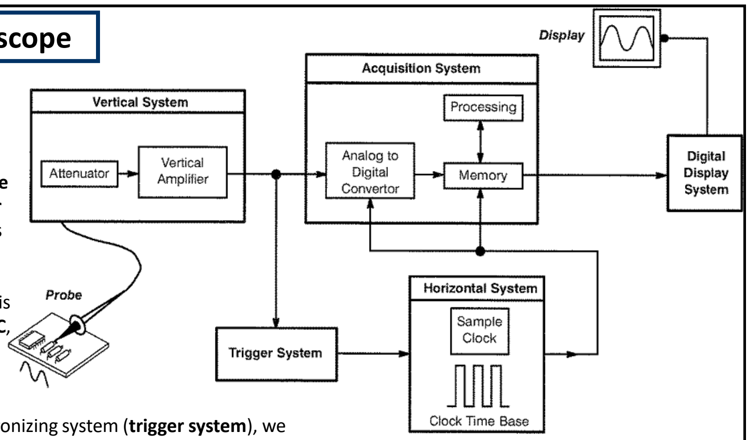
Digital oscilloscope

As with the analog oscilloscope, the input signal is divided or **amplified**. The resulting signal is **sampled with the help of an A/D converter (ADC)**, the digital signal is the **stored in memory**.

An important parameter is the **resolution of the ADC**, that is, in most oscilloscopes, **8 bits**.



With the help of a synchronizing system (**trigger system**), we synchronize the clock signal to the measured signal, e.g. we start sampling when **the signal exceeds a certain threshold with a certain slope**. From then on we record datapoints based on the memory size and the settings (**record length**), and the measured voltage points are displayed as a function of time elapsed since the trigger signal.

Sampling is controlled by a clock signal (**sample clock**). Its frequency gives the **sampling frequncy** (samples/second).

On a digital oscilloscope, even the part of the signal before the trigger event can be displayed, as the sampling is continuous, and the trigger only specifies, what part of the signal is displayed, and what part is ”thrown away”. This is called **pretrigger.**

Digital oscilloscopes can do further operations on the signal (e.g. **smoothing**, **averaging**).

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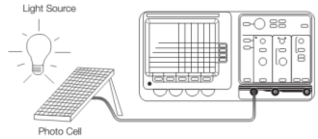
Basic settings

***Voltage division**: Volts/division*

***Vertical shift:** (vertical pos.)*

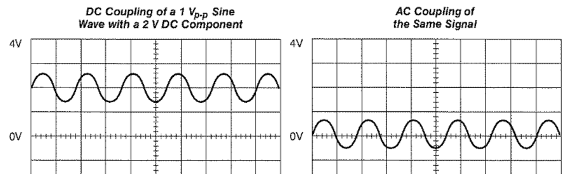
***Time base**: Seconds/division*

Experiments: measuring "blinking" light

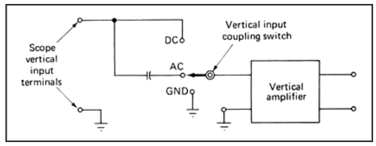


Input setup

DC/AC coupling; in case of GND – AC coupling, a capacitor is connected in series with the input, so the input sees only the alternating signal, the direct voltage (DC) is filtered out. GND grounds the input.



Termination: 50 Ω, 75 Ω, 1 MΩ – setting input resistance
Invert ON/OFF – optional, displays the signal × (-1)
Bandwidth FULL/20MHz – the input bandwith can be limited



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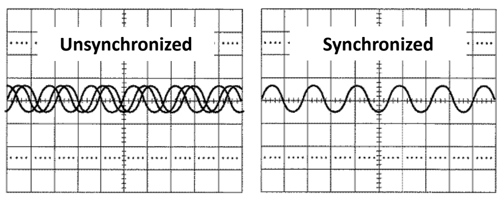
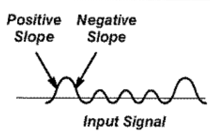
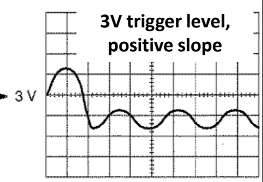
Trigger settings

Without synchronization (triggering), the waveforms will appear in different phases on the screen

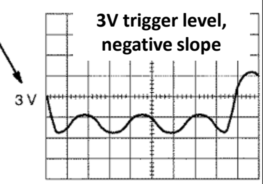


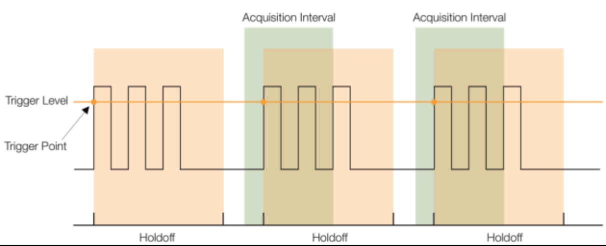
Source – which channel is used for triggering
Level – what voltage threshold to trigger to
Slope – trigger to positive or negative slope
Coupling – trigger coupling
(AC, DC, HF reject, LF reject, stb.)





Normal mode: records a curve after each trigger event
Auto mode: if it does not get a trigger signal, sampling is started automatically ("let us see something even if the trigger is uncertain")
Single shot: records the curve corresponding to the first trigger event, then stops (does not wait for more curves) Restart: **Run/Stop button**.





Holdoff: after the trigger event, it will not accept a new trigger event in the set holdoff time period.

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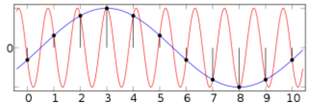
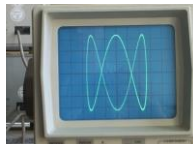
Acquire menu

Experiment: aliasing



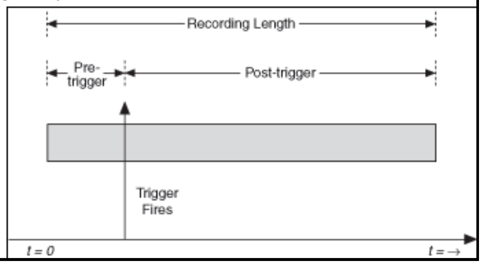
Mode:
Sample – the raw, sampled datapoints are displayed
Enhanced resolution – More neighboring datapoints are averaged together, thus the vertical voltage resolution is improved, but the horizontal, time resolution is decreased. Repeating signals are not needed for this.
Average – averages consecutive wavefroms, thus the voltage resolution is enhanced, and time-resolution is not decreased, however, longer waiting period is needed and can be used only with precisely repeating signals.
Etc.

Record length: *pl. 1k, 10k, 100k, 1M, 5M, 10M points* – how many points should a sampling take. For example: we want to resolve one period of a 50 Hz signal, so the sampling time-window is 20 ms. In case of 10M datapoints, the time-resolutions in 2 ns, thus, superimposed to the 50 Hz signal, even a few hundred MHz signal can be resolved. Processing a lot of datapoints can take long, especially if we do additional mathematical operations.



XY mode: The horizontal axis does not stand for time, rather we display the measured voltage as a function of the voltage of another channel. E.g. Lissajous curves

Trigger position, pretrigger: where the sampling window should be with respect to the trigger signal. We can record datapoints before the trigger (pretrigger)!



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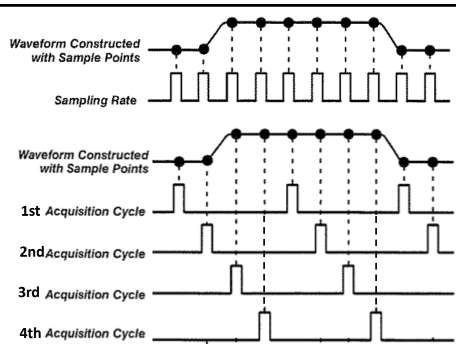
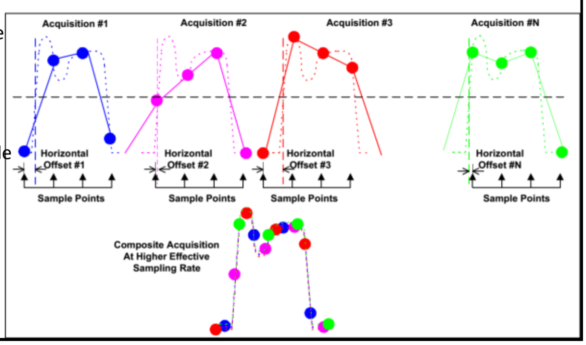
Sampling methods

Real time sampling: measurement points are taken at intervals corresponding to the clock signal. Even a single curve can be taken, no repetitive signal is needed.
Sequential equivalent time sampling: after a trigger event, a measurement point is taken with time delay Δt, and for successive trigger events, this Δt is always increased a little. Advantage: significantly higher time resolution can be achieved than with real time sampling. Disadvantages: can only be used for repetitive signals with a precise trigger event and no possibility to measure before the trigger event (pretrigger).



Random equivalent time sampling: the scope samples continuously, with trigger events occurring at random times relative to the sampling period (the measured signal and the clock signal are not synchronous). The time between the trigger and the first sampling after the trigger is measured much more accurately than the sampling period. Measuring the repetitive waveform at multiple trigger events and shifting the waveforms together with the measured trigger-sampling event delays the waveform can be resolved with a much better temporal resolution than with real time sampling. Pretriggering is also possible, but this method is slower than sequential equivalent time sampling.



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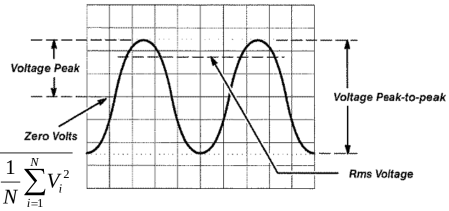
Measurements, math, cursors



Measurements: numerous measurements can be done with the recorded waveform, e.g. *period, frequency, mean voltage, RMS voltage, peak-to-peak voltage, maximum/minimum voltage, rise time/fall time,* etc.

Definition of RMS voltage for continuous and discrete signal:

$$V_{\mathrm{RMS}} = \sqrt{\frac{1}{T}\int_0^T [V(t)]^2 \mathrm{d}t}\,; \quad V_{\mathrm{RMS}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} V_i^2}$$

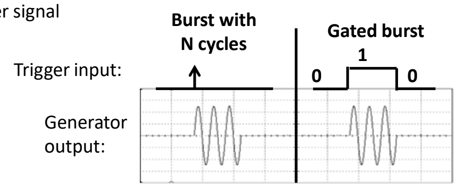
Illustratively: the DC voltage, whose power is the same as the average power of the alternating signal

Measurements: with the recorded waveform, or waveforms, mathematical operations can be done, e.g. *addition, multiplication, numerical differentiation, Fourier transformation*, etc.

Cursors: a horizontal and a vertical cursor pair can be placed on the screen, whose position, and the distance between them can be read.

Waveform generator (AWG, arbitrary waveform generator)

D/A converter with accurate clock signal, with that *sine, sawtooth and sqaure signs, pulses*, or even *arbitrary* programmed *waveforms* can be released. Frequency, amplitude, offset, symmetry, pulse width etc. can be set.
Advanced generators can be synchronized to external trigger signal (***burst mode***):



Burst with N cycles: gives out N periods from the signal after every trigger event
Gated burst: only gives out signal when trigger input is in state 1

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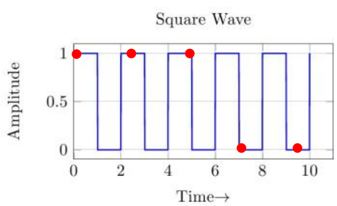
Experiment: Measuring „blinking” light

We want to detect the light of a distant tungsten lamp. The detector shows a large DC signal, whose amplitude hardly changes as the light is switched on and off. How can we improve the measurement? As the light bulb is „blinking” with a frequency of 100 Hz (if the voltage alternates with 50 Hz, the intesnity alternats with 100 Hz!), we can get rid of the DC background if we measure in AC coupling, and the signal can be amplified. Now the 100 Hz blinking shows, but very noisy, and it „crawls” horizontally on the screen. If we decrease the oscilloscope's bandwidth, (e.g. from 100 MHz to 20 MHz), the signal becomes nicer, because the high-frequency noise is filtered out. The same can be achieved with the „enhanced resolution” function, in which the scope averages consecutive datapoints. With that, time resolution worsens, but voltage resolution improves. The horizontal „crawling” can be removed with appropriate triggering, however, we cannot trigger to a noisy signal, because the voltage noise will bring in noise in the triggering time as well. Best practice is to trigger to an external signal with large amplitude, that is in phase with our measured signal. (Most function generators can emit such a trigger sigal, that is large-amplitude square sign in phase with the drive signal.) In our case we can trigger to the network (50 Hz) frequency, a function avaliable on most oscilloscopes. If we succeeded with appropriately triggering, then averaging the consecutive curves (average function) will get rid of the noise on our 100 Hz signal.

Experiment: Aliasing

We see a large-amplitude, noiseless square signal on the scope, yet we cannot trigger to it, no matter how well we set the trigger level. What is the reason? In reality, we measure a square signal with frequency 100,010 Hz, in a 300 ms wide time window, with a sampling rate of 0.3 ms (1000samples/curve). Meanwhile, the period of our signal is ~0.01 ms, meaning we record a datapoint only in every 30th period, but because of the 10 Hz „off-tuning”, this point is recorded from a later part of the signal, and a latent square wave forms (aliasing). This problem can be avoided if the sampling rate is set in accordance with the period of the signal.



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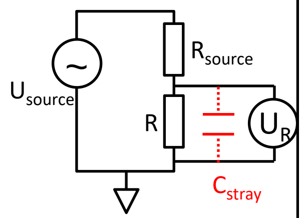
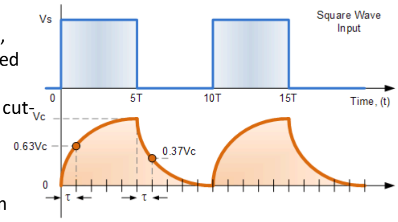
RC time constant

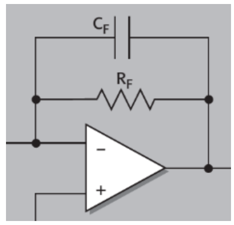


Cables and circuit elements always have a finite stray capacitance, which, together with the appropriate resistors in the circuit, forms an RC filter. Example: AC voltage source (U_{source}) is used to test resistance R in current generator mode. The circuit current is set with resistor R_{source}>>R (i.e. $I_{source} \approx U_{source}/R_{source}$), the voltage is measured at R. The resistor R and the long coax cable form a parallel RC circuit. Without the stray capacitance $U_R = I_{source}R$ voltage should drop on the resistor. However, if the absolute value of the stray capacitance's complex impedance becomes smaller than the resistance, $|Z_C|=1/\omega C_{stray}<R$, most of the current flows through the capacitance and the measured U_R voltage strongly decreases. This means, that at frequency $2\pi f=\omega<<1/RC_{stray}$ the desired voltage is measured, while above the cut-off frequency $\omega=1/RC_{stray}$ the measured voltage is cut-off, i.e. it becomes smaller as the frequency is increased.



The typical capacitance of a coax cable is 50-100pF/m, so with a 6m cable and R=3.3kΩ (see experiment) we expect a cut-off at f~160-320kHz.

If we drive the circuit with a square wave, we see a charge/discharge at the leading and trailing edges with a time constant of $\tau=RC_{stray}$



Another example: when measuring low currents, we usually use a current amplifier with a high feedback resistor. With R_F=1GΩ feedback resistor and C_F=1pF stray capacitance, the instrument bandwidth is $f_{max}=1/2\pi RC_F$~160Hz, you cannot measure a signal with a higher frequency than this, i.e. you cannot measure small currents quickly. A significantly lower stray capacitance than this is not very feasible, but the stray capacitance can be somewhat compensated by clever tricks.

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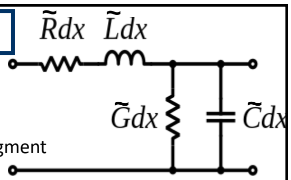
Signal propagation in coaxial lines, telegrapher's equations



If the length of the cable is comparable to the wavelength of the electromagnetic wave propagating at a given frequency, then the description for low frequency circuits is no longer applicable!

Coaxial cable model: the outer (shield) wire is grounded. A dx long elementary segment of the inner conductor has a self-induction coefficient $\tilde{L}\mathrm{d}x$ and a resistance $\tilde{R}\mathrm{d}x$. Furthermore, a dx long elementary segment of the inner conductor has a $\tilde{C}\mathrm{d}x$ capacitance and $\tilde{G}\mathrm{d}x$ conductance towards the shield. From theses parameters the resistance of the inner wire and the conductance towards the shield is neglected ($\tilde{R}=0, \tilde{G}=0$). Due to the self-inductance, the voltage drop on the elementary section of wire dx is:

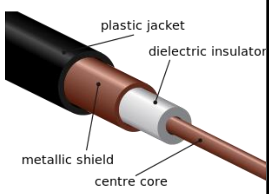
$$\mathrm{d}U=-\tilde{L}\mathrm{d}x\cdot\frac{\mathrm{d}I}{\mathrm{d}t},$$

$$\tilde{L}=\frac{\mu_0\mu_r\ln(D/d)}{2\pi},\quad \tilde{C}=\frac{2\pi\varepsilon_0\varepsilon_r}{\ln(D/d)}$$

D and d are the diameters of the outer and inner wires

while the variation of the current along a dx segment is determined by the charging and discharging of the capacitance towards the shield:

$$\mathrm{d}I=-\tilde{C}\mathrm{d}x\cdot\frac{\mathrm{d}U}{\mathrm{d}t}.$$



These can be used to write down the coupled differential equations on the current and voltage, the so-called telegrapher's equations:

$$\frac{\partial U(x,t)}{\partial x}=-\tilde{L}\cdot\frac{\partial I(x,t)}{\partial t},\quad \frac{\partial I(x,t)}{\partial x}=-\tilde{C}\cdot\frac{\partial U(x,t)}{\partial t},$$

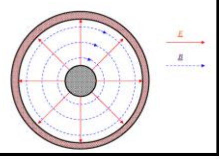
which simply give the following wave equations:

$$\frac{\partial^2 U(x,t)}{\partial t^2}=v_{\mathrm{signal}}^2\cdot\frac{\partial^2 U(x,t)}{\partial x^2},\qquad \frac{\partial^2 I(x,t)}{\partial t^2}=v_{\mathrm{signal}}^2\cdot\frac{\partial^2 I(x,t)}{\partial x^2},$$

where v_{signal} is the propagation speed of the EM wave in the coaxial cable:

$$v_{\mathrm{signal}}=\frac{1}{\sqrt{\tilde{L}\tilde{C}}}$$

the signal travels at approximately ~2/3 the speed of light in coaxial cables



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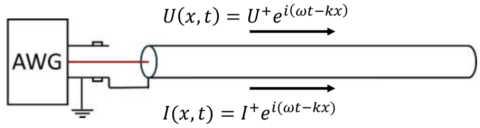
Solution of the telegrapher's equation for waves with ω circular frequency

First, let's assume that a wave with ω circular frequency travels in an infinite transmission line in positive direction. In this case no reflection (i.e. no wave travelling in the negative direction) is considered:

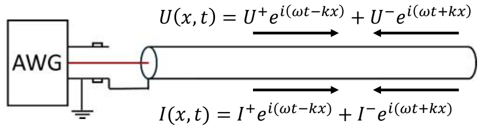


Substituting this to the first telegrapher's equation the relation between the voltage-wave and the current-wave is obtained:

$$-ikU^+ e^{i(\omega t - kx)} = -i\omega \tilde{L} I^+ e^{i(\omega t - kx)} \Rightarrow \frac{U^+}{I^+} = \underbrace{\tilde{L}\frac{\omega}{k}}_{v_{\text{signal}}=1/\sqrt{\tilde{L}\tilde{C}}} = \sqrt{\frac{\tilde{L}}{\tilde{C}}} = Z_0.$$

Z_0 is the so-called wave impedance. This quantity has the dimension of resistance describing the ratio of the voltage and current wave-amplitudes.

In the general case, both positive and negative direction waves propagate in the line:



Substituting these wavefunctions to the both telegrapher's equation at the x=0 position the following relations are obtained:

$$-kU^+ + kU^- = -\tilde{L}\omega(I^+ + I^-)$$
$$-kI^+ + kI^- = -\tilde{C}\omega(U^+ + U^-)$$

From these the ratio of the voltage and current amplitudes is obtained for both the positive and negative direction waves, mind the negative sign in the latter case:

$$I^+ = U^+/Z_0, \quad I^- = -U^-/Z_0$$

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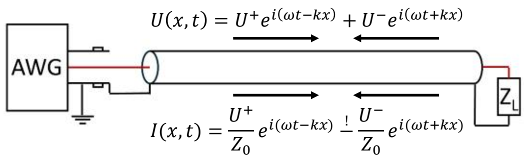
Reflections on a terminating resistance, impedance matching

Now, we connect the inner wire and the shield at the end of the coaxial line through a Z_L load impedance, which introduces reflections at the wire end (x=0 position). The arbitrary solutions for harmonic waves are shown on the figure, note that the previous results for the ratios of the current and voltage are already applied.



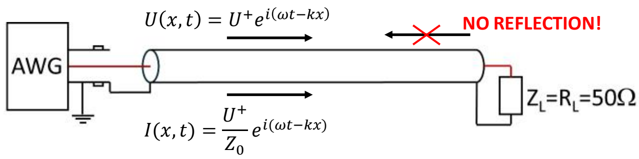
As a boundary condition, at the wire end the following equation should apply:

$$Z_L = \frac{U(x=0,t)}{I(x=0,t)} = \frac{U^+ + U^-}{\frac{U^+}{Z_0} - \frac{U^-}{Z_0}}$$

Rearranging this equation the ratio of the reflected voltage wave amplitude (U^-) and the incoming voltage wave amplitude (U^+) is obtained. This Γ coefficient describes the "reflectivity" of the Z_L terminating impedance:

$$\boxed{\Gamma = \frac{U^-}{U^+} = \frac{Z_L - Z_0}{Z_L + Z_0}}$$

Let us first assume, that the terminating impedance is the same as the wave-impedance, $Z_L = Z_0$. The most common coaxial wires in the lab have 50Ω wave impedance, i.e. we terminate the wire with a 50Ω resistor. In this case the reflection coefficient is zero, i.e. only the positive direction harmonic wave coming from the arbitrary waveform generator (AWG) travels in the wire. This is the case of impedance matching ($Z_L = Z_0$), which prevents reflections at the terminating resistor:



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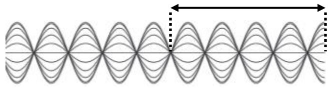
Reflections on a terminating resistance: free end

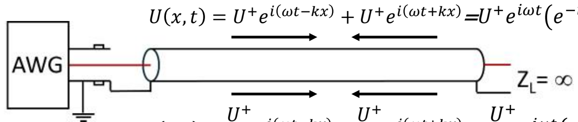
$$\Gamma = \frac{U^-}{U^+} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

Next, we assume, that the coaxial wire has a free end, i.e. the inner wire and the shield are disconnected ($Z_L=\infty$). In this case Γ=1 reflection coefficient is obtained, i.e the entire wave is reflected ($U^-=U^+$):

$$\left(n+\frac{1}{2}\right)\cdot\frac{\lambda}{2} = \left(n+\frac{1}{2}\right)\cdot\frac{\pi}{k} = \left(n+\frac{1}{2}\right)\cdot\frac{\pi v_{\text{signal}}}{\omega} = \left(n+\frac{1}{2}\right)\cdot v_{\text{signal}}/(2f)$$



$$U(x,t) = U^+e^{i(\omega t-kx)} + U^+e^{i(\omega t+kx)} = U^+e^{i\omega t}\left(e^{-ikx}+e^{ikx}\right) = \boldsymbol{2U^+e^{i\omega t}\cos(kx)}$$



$$I(x,t) = \frac{U^+}{Z_0}e^{i(\omega t-kx)} - \frac{U^+}{Z_0}e^{i(\omega t+kx)} = \frac{U^+}{Z_0}e^{i\omega t}\left(e^{-ikx}-e^{ikx}\right) = -\boldsymbol{\frac{2U^+}{Z_0}e^{i\omega t}\sin(kx)}$$

Note that here, the incoming and reflected voltage waves form a standing wave with a maximum voltage amplitude at the end of the wire, and cancellations at distances corresponding to half-integer multiples of the half wavelength. In contrast, the current-wave has cancellation at the wire end due to the infinite terminating impedance.

Note, that at low frequency ($kx \ll \pi/2$) the entire inner wire would oscillate according to the driving voltage ($2U^+e^{i\omega t}$). Note, that a factor of 2 difference is obtained if the function generator with 50Ω output resistance drives an impedance-matched (50Ω wave impedance) coaxial wire, or if it drives an infinite impedance system.

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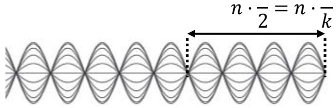
Reflections on a terminating resistance: short-circuit termination

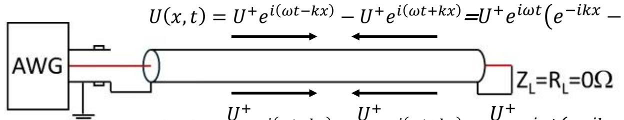
$$\Gamma = \frac{U^-}{U^+} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

Finally, we assume, that the coaxial wire-end is short-circuited (Z_L=0Ω). In this case Γ=-1 reflection coefficient is obtained, i.e the entire wave is reflected with a minus sign ($U^-=-U^+$):

$$n\cdot\frac{\lambda}{2} = n\cdot\frac{\pi}{k} = n\cdot\frac{\pi v_{\text{signal}}}{\omega} = n\cdot v_{\text{signal}}/(2f)$$



$$U(x,t) = U^+e^{i(\omega t-kx)} - U^+e^{i(\omega t+kx)} = U^+e^{i\omega t}\left(e^{-ikx}-e^{ikx}\right) = \boldsymbol{-2U^+e^{i\omega t}\sin(kx)}$$



$$I(x,t) = \frac{U^+}{Z_0}e^{i(\omega t-kx)} + \frac{U^+}{Z_0}e^{i(\omega t+kx)} = \frac{U^+}{Z_0}e^{i\omega t}\left(e^{-ikx}+e^{ikx}\right) = \boldsymbol{\frac{2U^+}{Z_0}e^{i\omega t}\cos(kx)}$$

Again, the incoming and reflected voltage waves form a standing wave, which has zero voltage at the wire end due to the short circuit, and cancellations are also observed at integer wavelength distances. Note, that here the current wave has maximum amplitude at the wire end.

Note, that at low frequency ($kx \ll \pi/2$) the entire inner wire would be at zero voltage due to the short circuit at the end.

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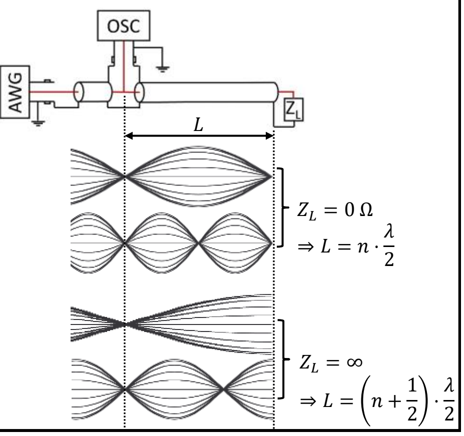
Experiments with transmission lines: measurement of propagation speed through standing wave tests

Experiment: testing cable-end reflection with harmonic excitation. The sinusoidal output of a signal generator is connected to the input of an oscilloscope by a short coaxial cable, and a long (L≈6m) coaxial cable is connected to the input of the oscilloscope by a T-divider, the end of which is first terminated with 50Ω. Changing the frequency (f) does not change the amplitude of the signal measured by the oscilloscope in this case.

If the far end of the long coax is terminated with zero resistance (short circuit), oscillations are seen as a function of frequency with dedicated cancellations of the signal. The lowest frequency cancellation corresponds to a standing wave of a half wavelength on the coax cable. Further cancellations are observed at $L = n \cdot \frac{\lambda}{2}$, where n is an integer.

If the far end of the long coax is terminated with infinite resistance (free end), similar oscillations are seen as the frequency is varied, but in this case the lowest frequency cancellation at the oscilloscope corresponds to a standing wave of a quarter wavelength, and further cancellations are observed at $L = \left(n + \frac{1}{2}\right) \cdot \frac{\lambda}{2}$.

Knowing the f frequency of the cancellations and the above relations between cable length and wavelength at the cancellation points the speed of the signal propagation can be determined from the $v_{\text{signal}} = \lambda \cdot f$ relation.



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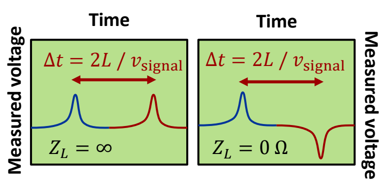
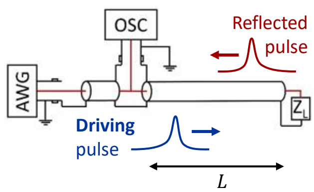
Experiments with transmission lines: measurement of propagation speed by measuring reflected pulses

Experiment: testing cable-end reflection using pulses. Instead of sinusoidal driving we use pulses. The output pulse is seen directly at the input of the oscilloscope, but the pulse reflected from the end of the cable is also displayed with a time-delay. In the case of a 50Ω termination there is no reflected signal. In case of free wire-end ($Z_L = \infty$) the entire pulse is reflected with identical sign to the original pulse. In case of short circuit at the wire end (Z_L=0Ω) the entire pulse is reflected with opposite sign. From the time difference between the original and the reflected pulse, the propagation speed of the signal (~2/3 speed of light) can be determined.

Note, that the $\Gamma = \frac{U^-}{U^+} = \frac{Z_L - Z_0}{Z_L + Z_0}$ formula was derived for harmonic waves. A pulse can be decomposed to harmonic waves with various circular frequencies ("Fourier transformation", see later), i.e. the above formula applies for the harmonic wave components. If the terminating impedance is a simple resistor ($Z_L = R_L$), the above formula is independent of the circular frequency, i.e. the reflected pulse does not change its shape, and the Γ reflection coefficient is valid for the pulse amplitude. For capacitive and inductive loads, however, the frequency-dependent reflection coefficients yield the strong modification of the pulse shape.





Finding contact faults in the middle of the wire: while a portion of the pulse still travels to the wire end to be reflected, partial reflections also occur at the position of the fault. The distance of the fault can be determined by the time-delay of the partial wave reflected on the fault.

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Experiments with transmission lines: the case of non-coaxial cables

Gedanken-experiment: a high-frequency signal is passed through a simple audio wall cable, or in an old flat antenna cable (see figures). The transmission of the cable is tested as a function of frequency using an oscilloscope. If the cable is placed in water, the measured voltage changes significantly. This is because the dielectric constant of the medium between the two wires changes significantly, i.e. the capacitance between the wires changes. Such a phenomenon is not observed with coaxial cables, where the capacitance between the inner core and the shielding is determined by the fixed geometry and the dielectric constant of the plastic insulation surrounding the inner core, and is not significantly affected by environmental influences.

Furthermore, in a rolled old antenna cable the signal at the beginning of the cable can be directly coupled to the cable end without propagation through the entire cable length. This phenomenon also yields unwanted signal distortion.

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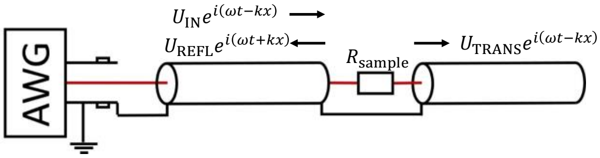
Transmission and reflection on an inline resistor



What happens if the resistor is not placed at the wire-end, but between the inner wires of two coaxial cables. An incoming harmonic wave with U_{IN} amplitude is sent to the resistor from the AWG. Due to the wave-nature of the signal propagation, a part of the wave is reflected on the resistor (U_{REFL}), while the other part is transmitted (U_{TRANS})

$$U_{\mathrm{LEFT}} = U_{\mathrm{IN}}e^{i(\omega t-kx)} + U_{\mathrm{REFL}}e^{i(\omega t+kx)},\quad U_{\mathrm{RIGHT}} = U_{\mathrm{TRANS}}e^{i(\omega t-kx)}$$

$$I_{\mathrm{LEFT}} = \frac{U_{\mathrm{IN}}}{Z_0}e^{i(\omega t-kx)} - \frac{U_{\mathrm{REFL}}}{Z_0}e^{i(\omega t+kx)},\quad I_{\mathrm{RIGHT}} = \frac{U_{\mathrm{TRANS}}}{Z_0}e^{i(\omega t-kx)}$$

Current conservation at the resistor (x=0 position): $I_{\mathrm{LEFT},x=0} = I_{\mathrm{RIGHT},x=0} \Rightarrow U_{\mathrm{IN}} - U_{\mathrm{REFL}} = U_{\mathrm{TRANS}}$

Furthermore, we apply Ohm's law for the resistor:

$$U_{\mathrm{LEFT},x=0} - U_{\mathrm{RIGHT},x=0} = I_{\mathrm{LEFT},x=0} \cdot R_{\mathrm{sample}}$$
$$\Rightarrow U_{\mathrm{IN}} + U_{\mathrm{REFL}} - U_{\mathrm{TRANS}} = \frac{R_{\mathrm{sample}} \cdot U_{\mathrm{TRANS}}}{Z_0}$$

Merging these results (current conservation + Ohm's law): $\frac{U_{\mathrm{TRANS}}}{U_{\mathrm{REFL}}} = \frac{2Z_0}{R_{\mathrm{sample}}}$

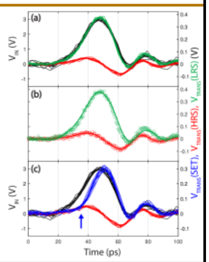
Or alternatively:

$$\frac{U_{TRANS}}{U_{IN}} = \frac{2Z_0}{R_{sample}+2Z_0},\qquad \frac{U_{\mathrm{REFL}}}{U_{\mathrm{IN}}} = \frac{R_{\mathrm{sample}}}{R_{\mathrm{sample}}+2Z_0}$$

The voltage drop on the sample:

$$U_{\mathrm{sample}}(t) = 2U_{\mathrm{REFL}}e^{i\omega t} = 2U_{\mathrm{IN}}e^{i\omega t}\frac{R_{\mathrm{sample}}}{R_{\mathrm{sample}} + 2Z_0}$$

Example: world-record resistive switching in memristors: https://doi.org/10.1002/aelm.202201104



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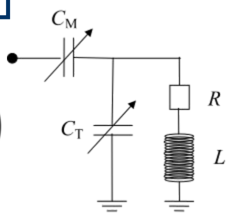
Complex impedance matching at a specific frequency



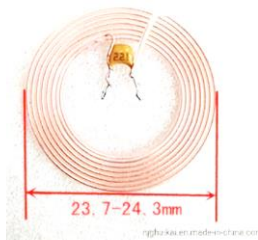
C_T sets the resonance frequency: $\omega_{res} \approx 1/\sqrt{LC_T}$

This scales the parasitic resistance up: $\mathrm{Re}\, Z_L(\omega = \omega_{res}) \approx R\left(1 + \frac{L/C_T}{R^2}\right)$

By proper choice of L and C_T, the real part of the impedance can be set to 50 Ohm

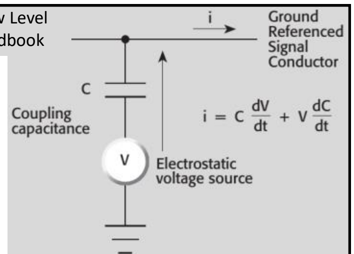
The imaginary part is inductive, can be compensated by C_M

Anti-theft LC circuit, inductive coupling

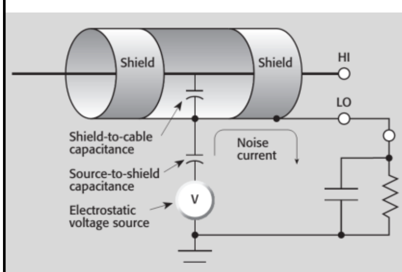


Suppressing external disturbances: electrostatic coupling, grounding

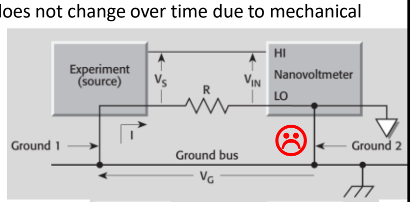
Source: Keithley Low Level Measurements Handbook



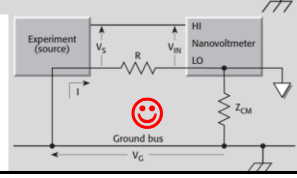
Our test leads can be capacitively coupled to external voltage sources. If the voltage or the capacitance varies with time (wires moving relative to each other), parasitic currents are generated in the circuit under test, which disturb the measurement. Causes: nearby power lines, RF fields, or even researchers walking past the measurement system :-). This causes unstable measurements, especially in high impedance circuits, where the accumulated charges are slowly discharged.



Solution: shield the circuit with a grounded metal enclosure or use shielded conductors. In this case, the currents generated by the electrostatic coupling flow through the shielding to the ground. It is also important that the cables are mechanically fixed, i.e. that the capacitance does not change over time due to mechanical vibrations.



Poor earthing can also cause problems, as there can be a voltage drop between different earth points, resulting in unwanted current flowing through the earth line. Solution: the circuit is grounded at one point (e.g. negative point of voltage source) and the other connections of the earth line (e.g. negative point of voltmeter) are floating or connected to ground through a finite resistor. The use of solid (low resistance, strongly connected) and short earth conductors is recommended.



Experiment – Shielding

A 1m BNC coax cable is connected to the input of the scope. We see zero voltage with very little noise. However, if this cable is supplemented with a 1m unshielded cable, we pick up considerable noise. (The shielding of the coax cable in the previous case solved the problem of filtering out this noise.) If the unshielded cable is inserted into a metal box, but the metal box is not grounded, we still see considerable noise, since the potential of the metal box can fluctuate and is transmitted to the cable. If we ground the box directly to the shielding of the coax cable, the noise is eliminated. However, if you ground the box to a completely different point on a long cable, the noise is only reduced, but not eliminated, as there can be a significant voltage drop on the long ground cable.

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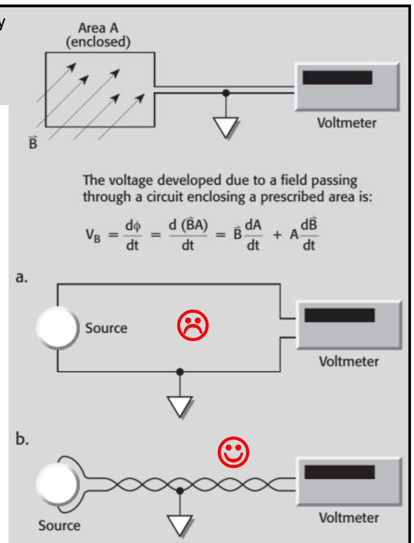
Suppressing external disturbances: Inductive coupling, twisted pairs

Source: Keithley Low Level Measurements Handbook



An unwanted voltage is induced due to a change in the magnetic flux in the area enclosed by the circuit. This can be caused by a changing magnetic field or even by the mechanical movement of the test wires in a static field. This is particularly a problem in low impedance circuits. Solution: mechanical clamping of wires and minimizing the flux around them (e.g. twisted pair wires where opposite fluxes occur at adjacent twists, so that the induced voltages on adjacent sections cancel each other out.)
Example: UTP wire = unshielded twisted pair.
Note: the slowly changing magnetic field is difficult to shield, so a material with a high magnetic permeability (e.g. mu-metal) is used. (At low temperature, the magnetic field is efficiently shielded by a superconductor enclosure.

High frequency disturbance, rectification

Due to the inductive and capacitive coupling, high frequency signals can also cause interference if they are not properly shielded (e.g. radio and TV transmissions, mobile phone WIFI, etc.). These appear as noise at the broadcast frequency (and its harmonics) but can also result in DC voltage in non-linear circuits. Above 100kHz electrostatic shielding is sufficient, no magnetic shielding is needed. But in case of nonlinear current-voltage characteristics, a radio frequency signal $V_{RF}\sin(\omega_{RF}t)$ "superimposed" on the V_{DC} measurement voltage induces a DC component due to the rectification:

Rectification (DC component!) Second harmonic generation

$$I\left(V_{DC}+V_{RF}\cos(\omega_{RF}t)\right)\approx I(V_{DC})+\left.\frac{dI}{dV}\right|_{V_{DC}} V_{RF}\cos(\omega_{RF}t)+\frac{1}{2}\left.\frac{d^2I}{dV^2}\right|_{V_{DC}} V_{RF}^2\cdot\overbrace{\cos^2(\omega_{RF}t)}^{(1+\cos(2\omega_{RF}t))/2}$$

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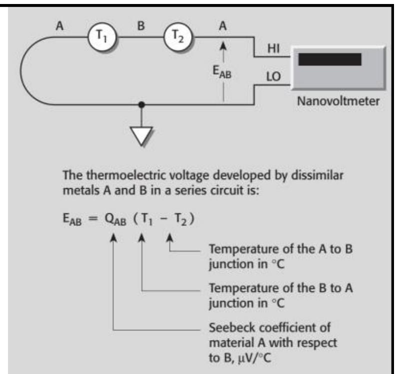
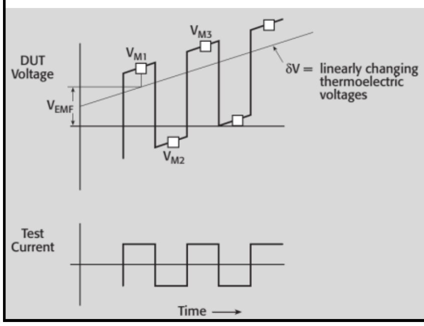
Suppressing external disturbances: thermoelectric voltages, offset compensation

When different metals come into contact, contact potentials occur, which cancel each other out in a closed circuit at a constant temperature. However, if the two contact points of metals A and B shown in the diagram are at different temperatures, a thermoelectric voltage will be induced in the circuit (this is exploited in thermocouple thermometers). At low signal levels, and with significant temperature gradients, the thermoelectric voltage can be compared to the measured signal.





Solution:
-ensure that the contact point pairs of the different metals are at the same temperature
-by varying the polarity of the driving current, the thermoelectric voltage can be compensated, since it does not depend on the current, while the voltage across the resistance under test varies in proportion to the current. The latter is the so-called offset-compensation method.

Source: Keithley Low Level Measurements Handbook

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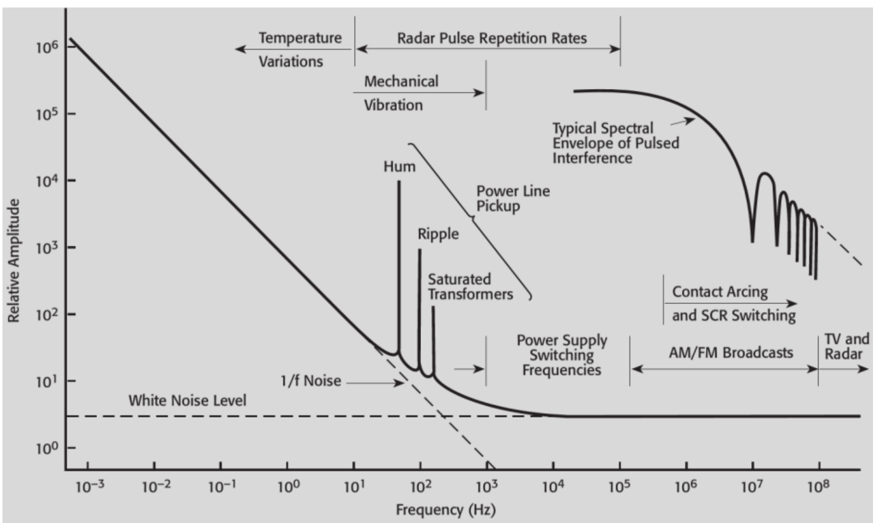
Typical frequency spectrum of external disturbance signals



Source: Keithley Low Level Measurements Handbook

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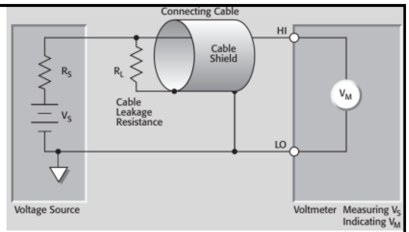
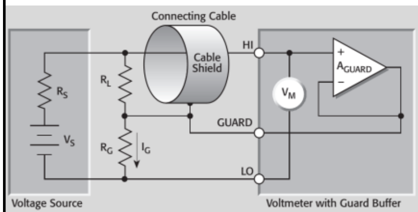
Guarding

Source: Keithley Low Level Measurements Handbook

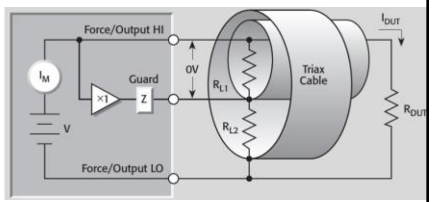
When measuring particularly high impedance circuits, the leakage resistance between the two wires (e.g. inner conductor and shield) can be comparable to the resistance of the circuit, causing significant measurement error.

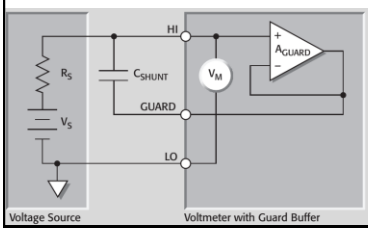




Solution: use a shielded wire, but do not ground the shielding, rather keep it at the potential of the inner wire using a 1x follower amplifier. This way there is no potential difference between the inner core and the shielding, i.e. the leakage current is suppressed. This is called *guarding*.

An even better solution is to use a triaxial cable, i.e. two concentric shields, the inner shielding follows the potential of the middle wire (guarding), while the outer shielding is grounded.





In a high output impedance circuit, the stray capacitance of the cable is also a problem, as the huge RC time constant means that it takes a very long time to reach a steady state VM value after connecting or changing the circuit. This can also be eliminated by guarding, since if the shielding is held at the potential of the inner core, the stray capacitance is not charged, i.e. the circuit responds much faster.

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